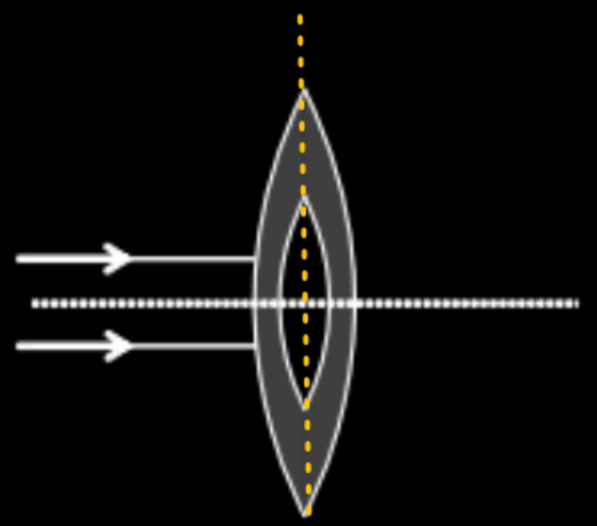
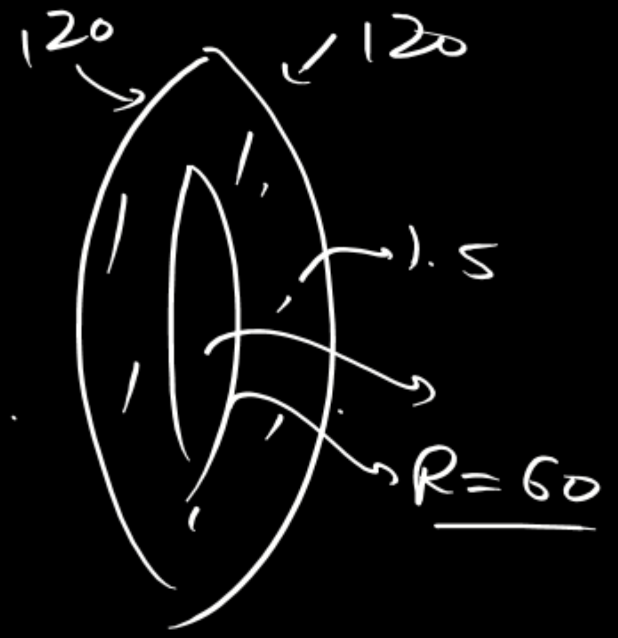
Figure shows an equi-convex lens with a cavity filled with air. The radius of curvature of both outer faces is 120 cm. The refractive index material of the lens is 1.5. The cavity has same shape as that of the lens but its linear dimensions are half of the lens. For the rays shown in the figure this will behave as –



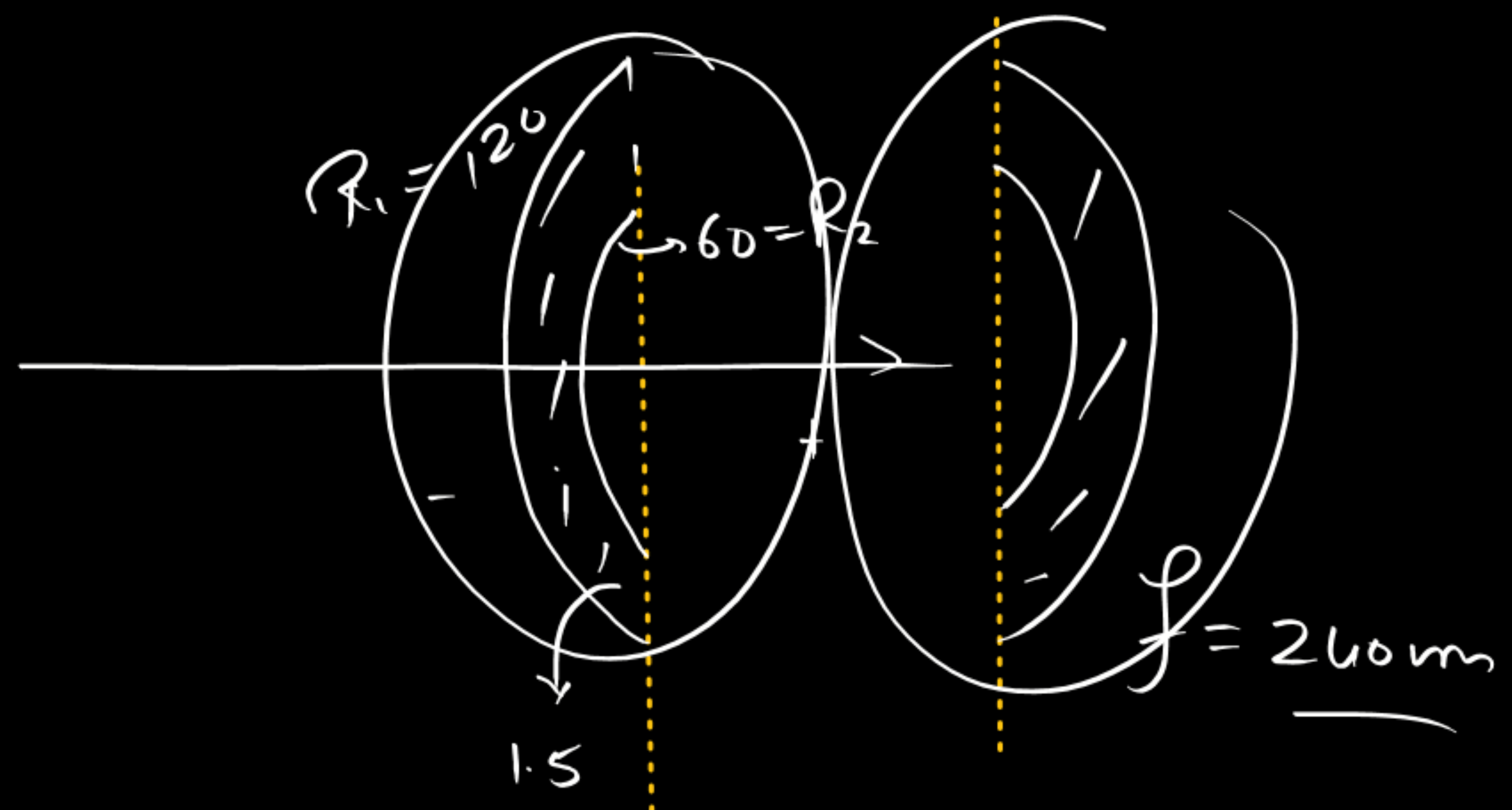


(A) Diverging lens of focal length 120 cm

(B) Converging lens of focal length 120 cm

(C) Lens of zero power

(D) Diverging lens of focal length 240 cm



$$\frac{1}{f} = (1.5-1)\left(\frac{1}{120} - \frac{1}{60}\right)$$

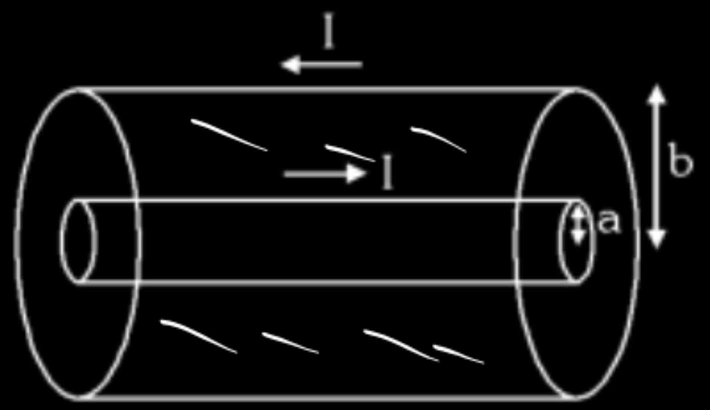
$$= \frac{1}{2} \times -\frac{1}{120}$$

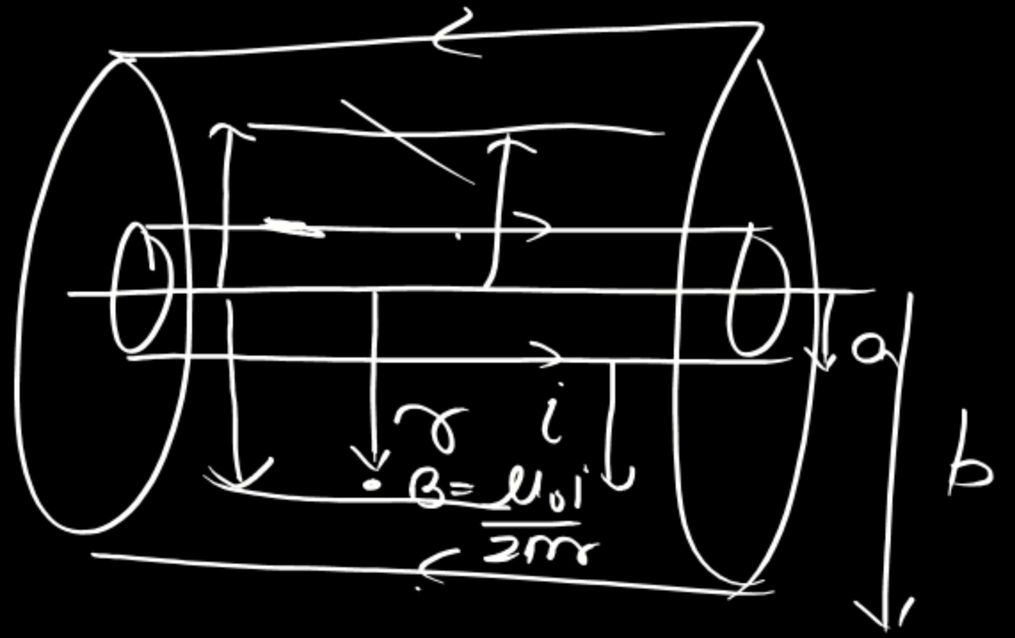
$$\boxed{f = -240\,\text{cm}}$$

$$\frac{1}{f_{eq}} = \frac{1}{-240} + \frac{1}{-240}$$

$$\boxed{f_{eq} = -120\,\text{cm}}$$

A long coaxial cable carries current I (along its surface). The current flows down the surface of inner cylinder of radius a and back along the outer cylinder of radius b.





(A) Energy stored in magnetic field per unit length is $\frac{\mu_0 I^2}{8\pi} \ln \frac{b}{a}$

(B) Energy stored in magnetic field per unit length is $\frac{\mu_0 I^2}{2\pi} \ln \frac{b}{a}$

(C) Self inductance of given arrangement per unit length is $\frac{\mu_0}{2\pi} \ln \left(\frac{b}{a}\right)$

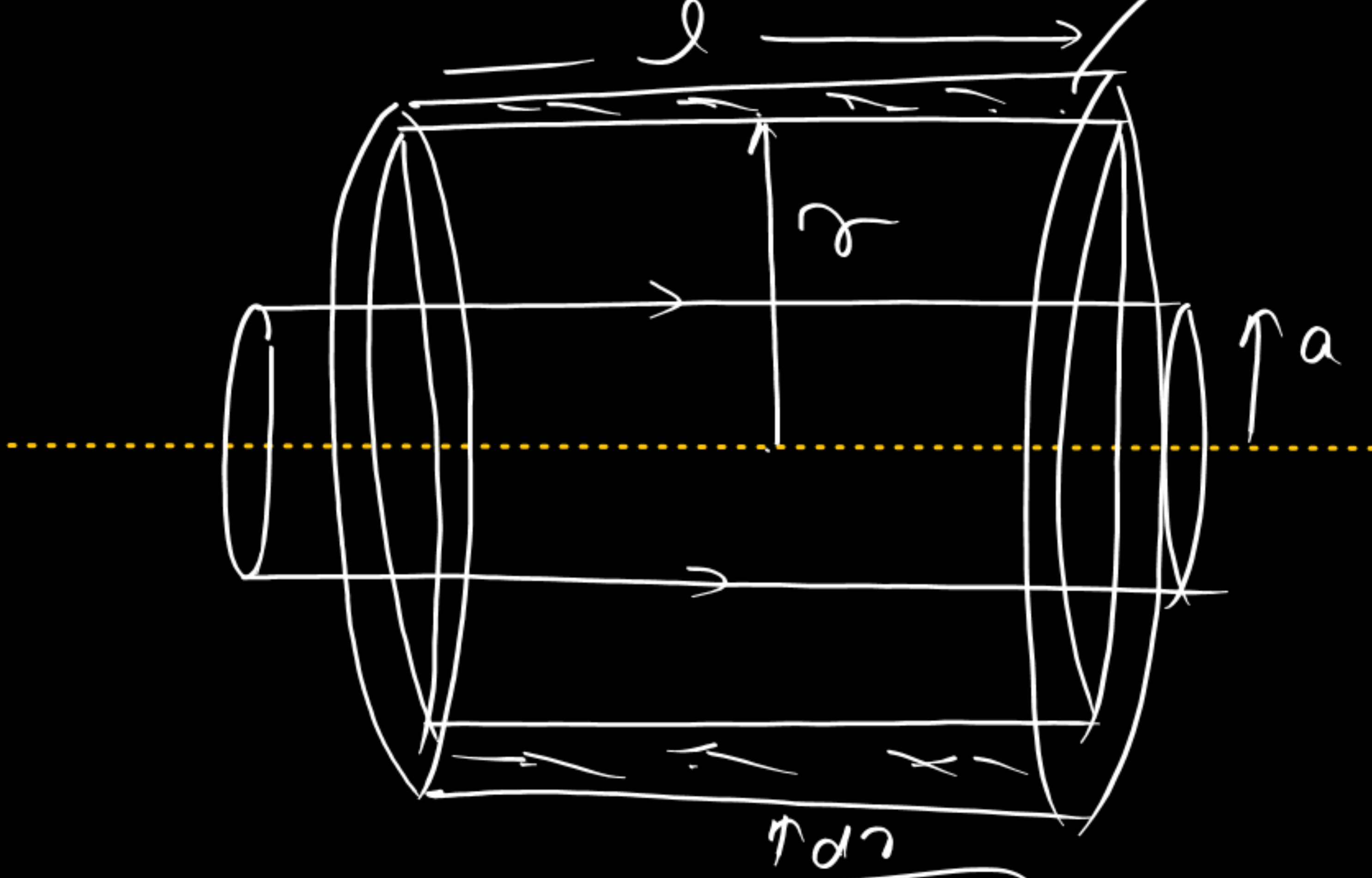
(D) Self inductance of given arrangement per unit length is $\frac{\mu_0}{4\pi} \ln \left(\frac{b}{a}\right)$

$\frac{B^2}{2\mu_0}$

$U = \frac{1}{2} L i^2$

$\frac{U}{l} = \frac{1}{2} \left(\frac{L}{l}\right) i^2$

$= \frac{1}{2} L' i^2$



$$dU = \frac{B^2}{2\mu_0}\, dV$$

$$= \frac{\left(\frac{\mu_0 I}{2\pi r}\right)^2}{\cancel{2}\mu_0} \times \cancel{2}\pi r l\, dr$$

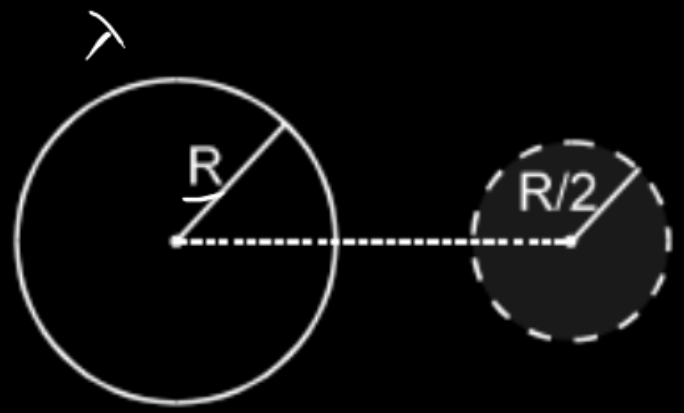
$$= \frac{\mu_0 I^2}{4\pi r}\, l\, dr$$

$$U = \int dU = \frac{\mu_0 I^2 l}{4\pi} \int_a^b \frac{dr}{r}$$

$$\boxed{\frac{U}{l} = \frac{\mu_0 I^2}{4\pi} \ln\frac{b}{a}} = \frac{1}{2} L' I^2$$

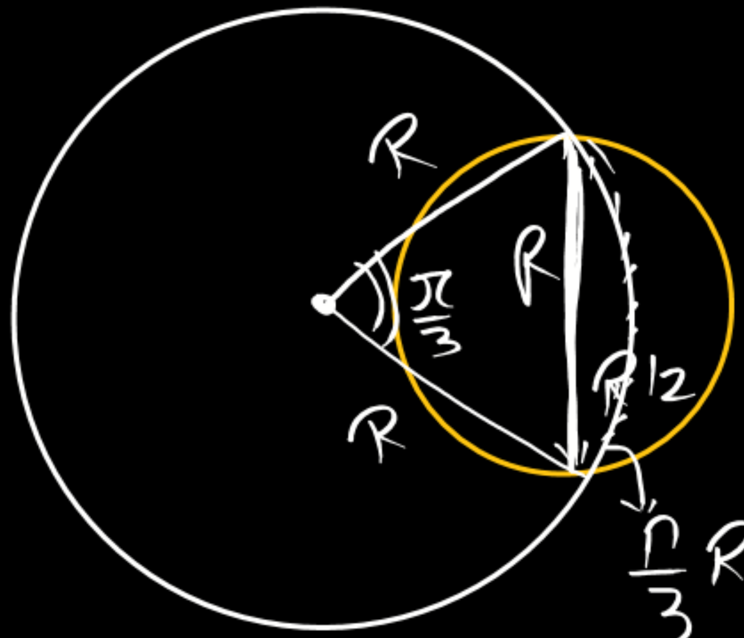
$$\boxed{L' = \frac{\mu_0}{2\pi} \ln\frac{b}{a}}$$

A ring of radius R having a linear charge density λ moves towards a solid imaginary sphere of radius $\frac{R}{2}$, so that the centre of ring passes through the centre of sphere. The axis of the ring is perpendicular to the line joining the centres of the ring and the sphere. The maximum flux through the sphere in this process is :



(A) $\frac{\lambda R}{\epsilon_0}$ (B) $\frac{\lambda R}{2\epsilon_0}$ (C) $\frac{\lambda \pi R}{4\epsilon_0}$ (D) $\frac{\lambda \pi R}{3\epsilon_0}$

$$\phi = \frac{q_{in}}{\varepsilon_0}$$



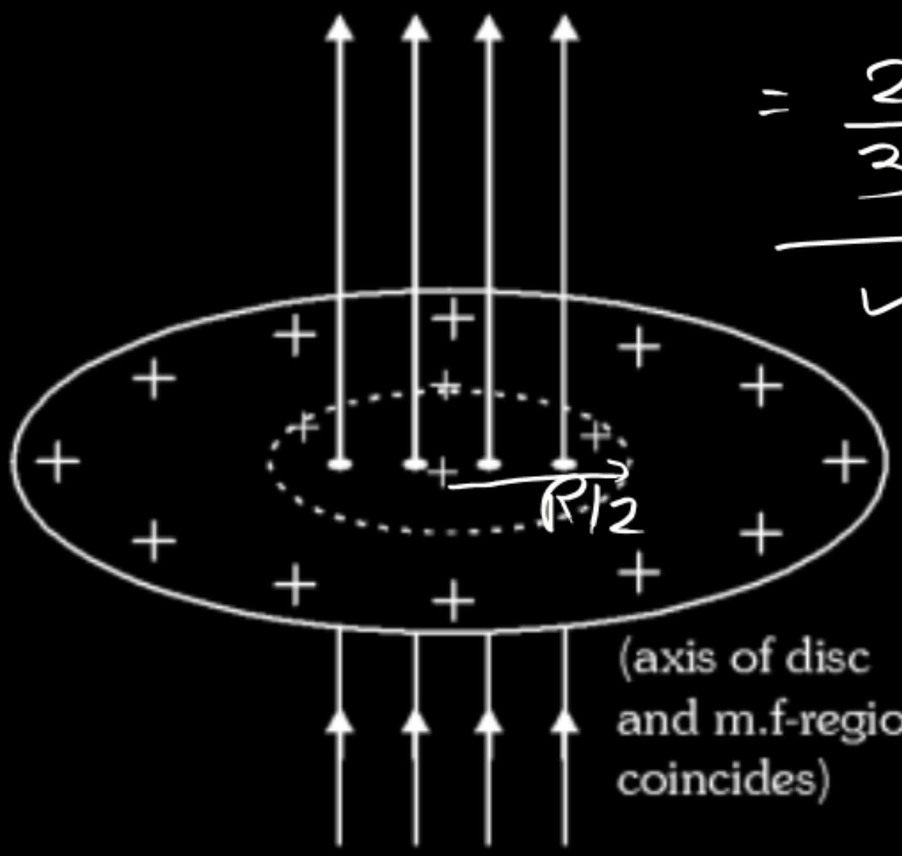
$$\frac{\frac{\pi}{3} R \lambda}{\varepsilon_0}$$

Uniform magnetic field $B = 2t^2$ Tesla present in cylindrical region of radius R/2. And a uniformly charged disc charge Q, mass m and radius R kept on rough horizontal plane with coefficient of friction μ. Find the time after disc start rotating.

$$B = 2t^2$$

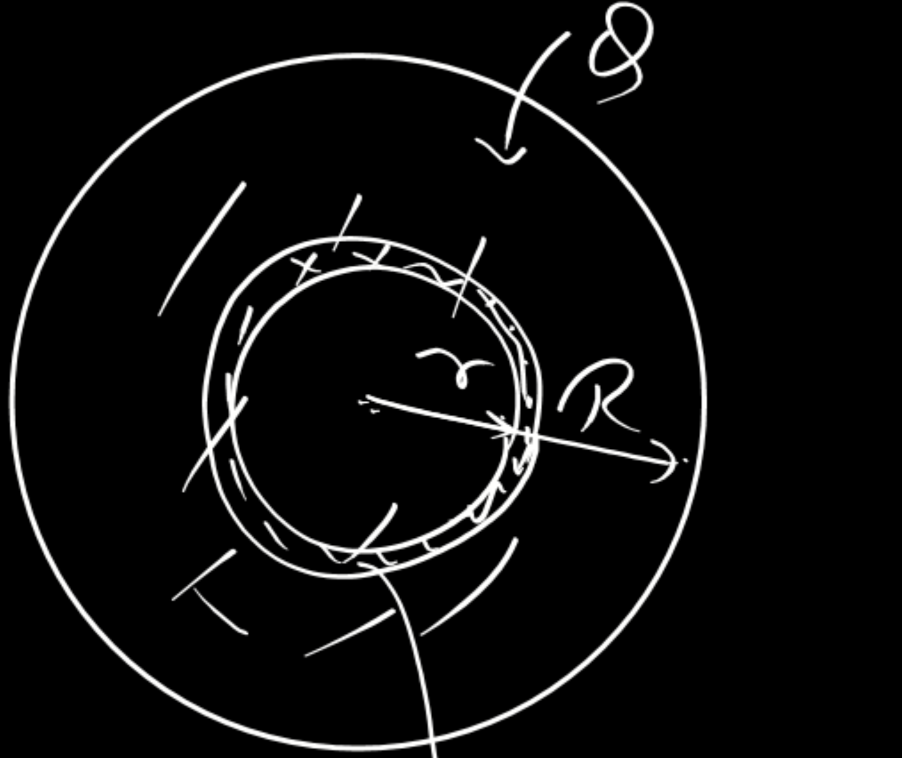
$$E_{in} = \frac{r}{2} \times 4t = 2rt$$

$$E_{out} = \frac{(R/2)^2}{2r} \times 4t = \frac{R^2 t}{2r}$$



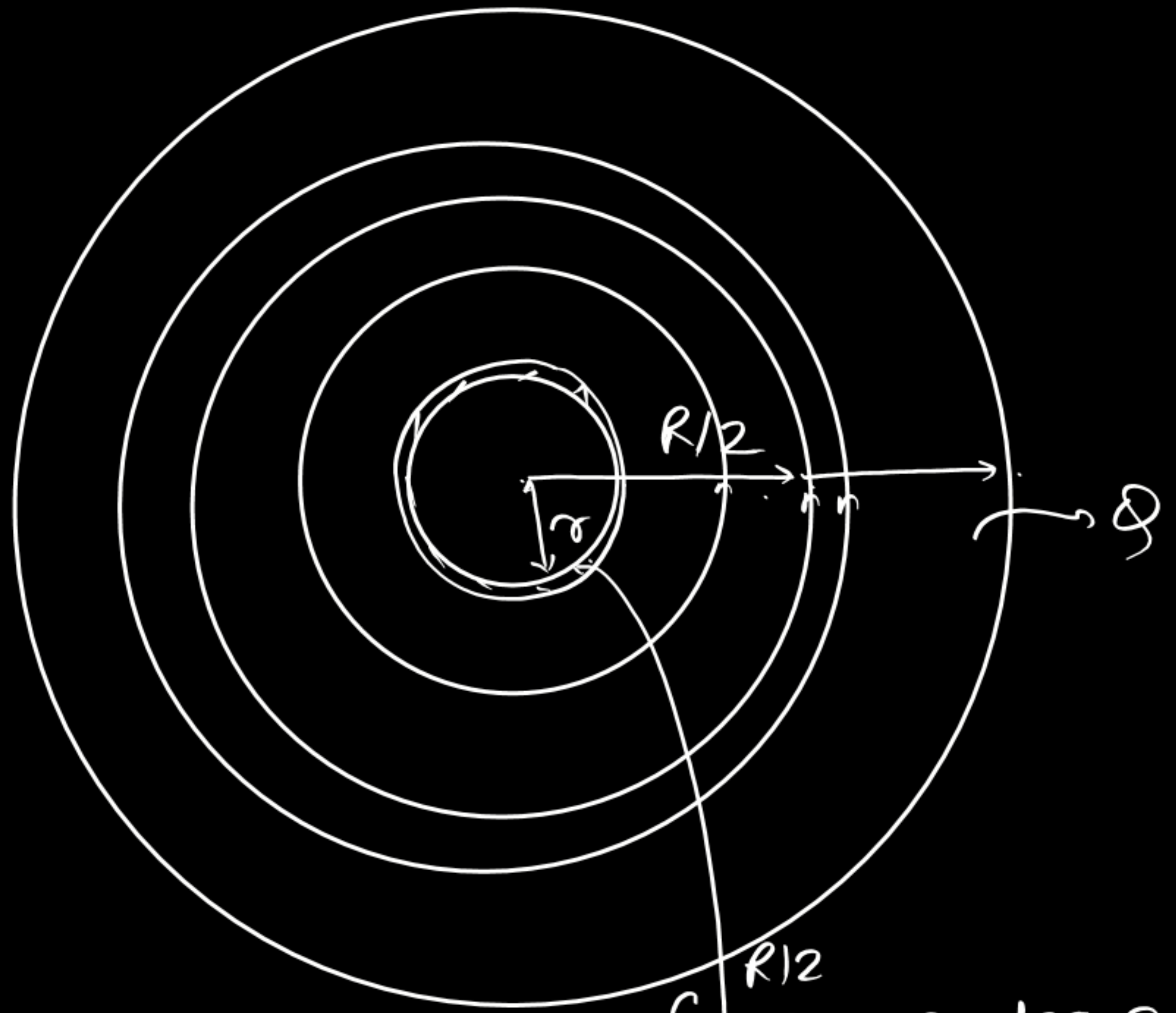
(axis of disc and m.f-region coincides)

$$\tau_f = \frac{2\mu mg}{R^2}\frac{R^3}{3} = \frac{2}{3}\mu mgR$$



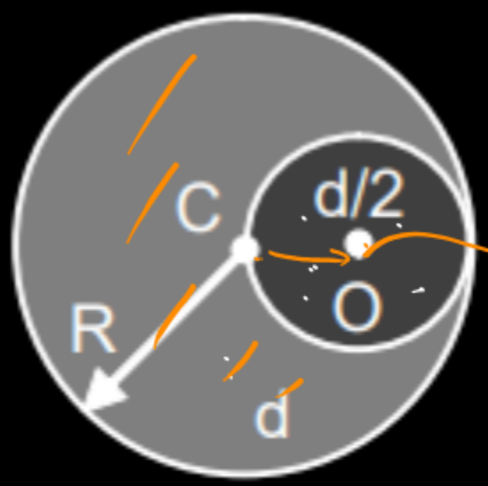
$$\tau = \int_0^R \frac{\mu mg}{\pi R^2} 2\pi r^2 dr$$

(A) $\frac{20\mu mg}{3R}$ (B) $\frac{32\mu mg}{27R}$ (C) $\frac{32\mu mg}{3R}$ (D) None of these



$$\int_0^{R/2} \frac{Q}{\pi R^2}\times 2\pi r\,dr\; 2rt\cdot r + \int_{R/2}^{R}\left(\frac{Q}{\pi R^2}\times 2\pi r\,dr\right)\frac{R^2 t}{2t}\cdot r$$

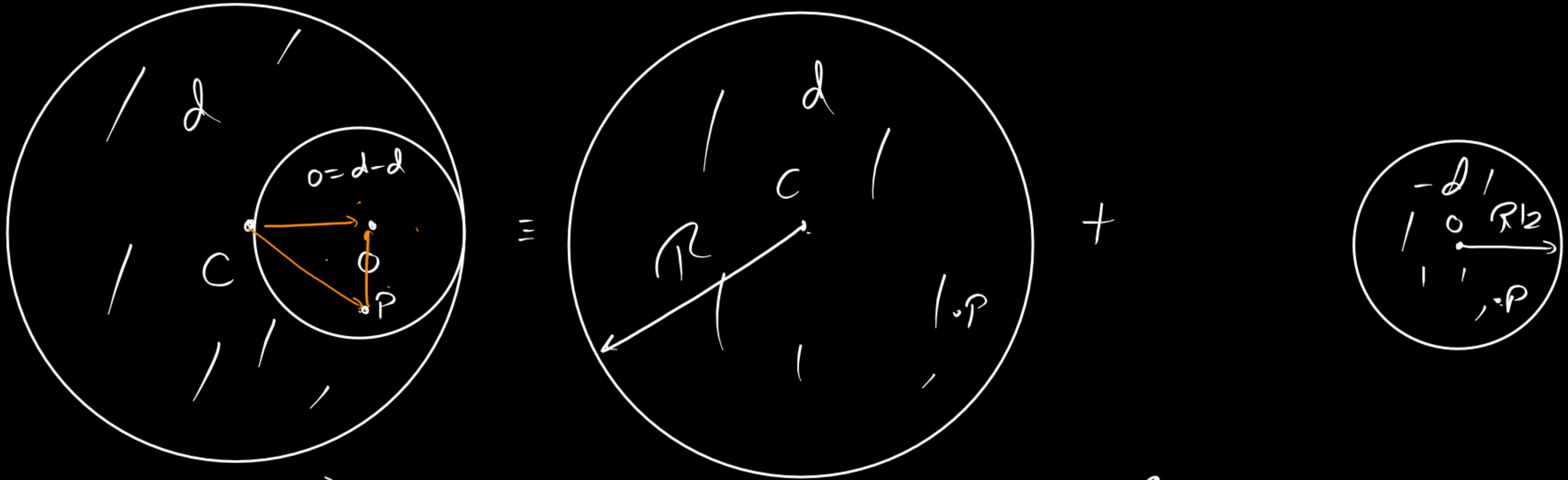
A fixed sphere of radius R and uniform mass density 'd' has a cavity of radius $\frac{R}{2}$ as shown in figure.



$$\frac{\cancel{\rho}d}{3\cancel{\varepsilon_0}} \times \frac{R}{2} \times 4\pi G = \frac{2\pi G R d}{3}$$

The centre of the solid sphere is at 'C' while that of the cavity is at 'O'. A liquid of density $\frac{d}{2}$ is filled in the cavity. The gravitational force exerted by the liquid on the solid sphere is

(A) $\frac{G\pi^2R^4d^2}{6}$ (B) $\frac{G\pi^2R^4d^2}{12}$ (C) $\frac{G\pi^2R^4d^2}{9}$ (D) $\frac{G\pi^2R^4d^2}{18}$ ✓

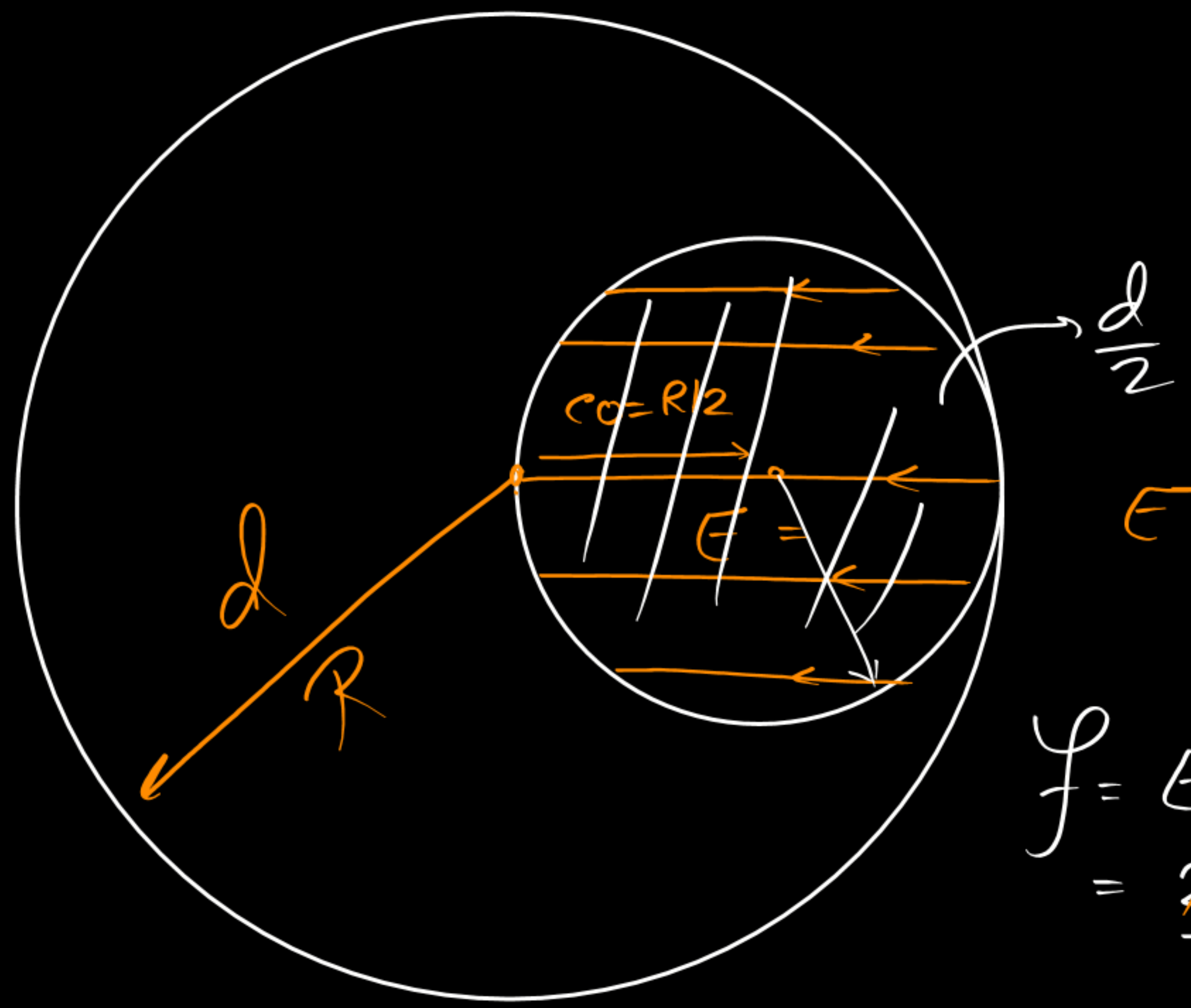


$$\vec{E}_g = \frac{-GM\vec{r}}{R^3}$$

$$E = -\frac{G\, d\, 4\pi \cancel{R^3}\, \vec{CP}}{3\cancel{R^3}} + \frac{G\, d\, 4\pi R^3\, \vec{OP}}{3\times 8 \times R^3/8}$$

$$= -\frac{G4\pi d}{3}\left\{\vec{CP} - \vec{OP}\right\} = -\frac{G4\pi d}{3}\left\{\vec{CP} + \vec{PO}\right\}$$

$$= -\frac{G4\pi d}{3}\vec{CO}$$



$$E = \frac{4\pi d G}{3} \times \frac{R}{2}$$

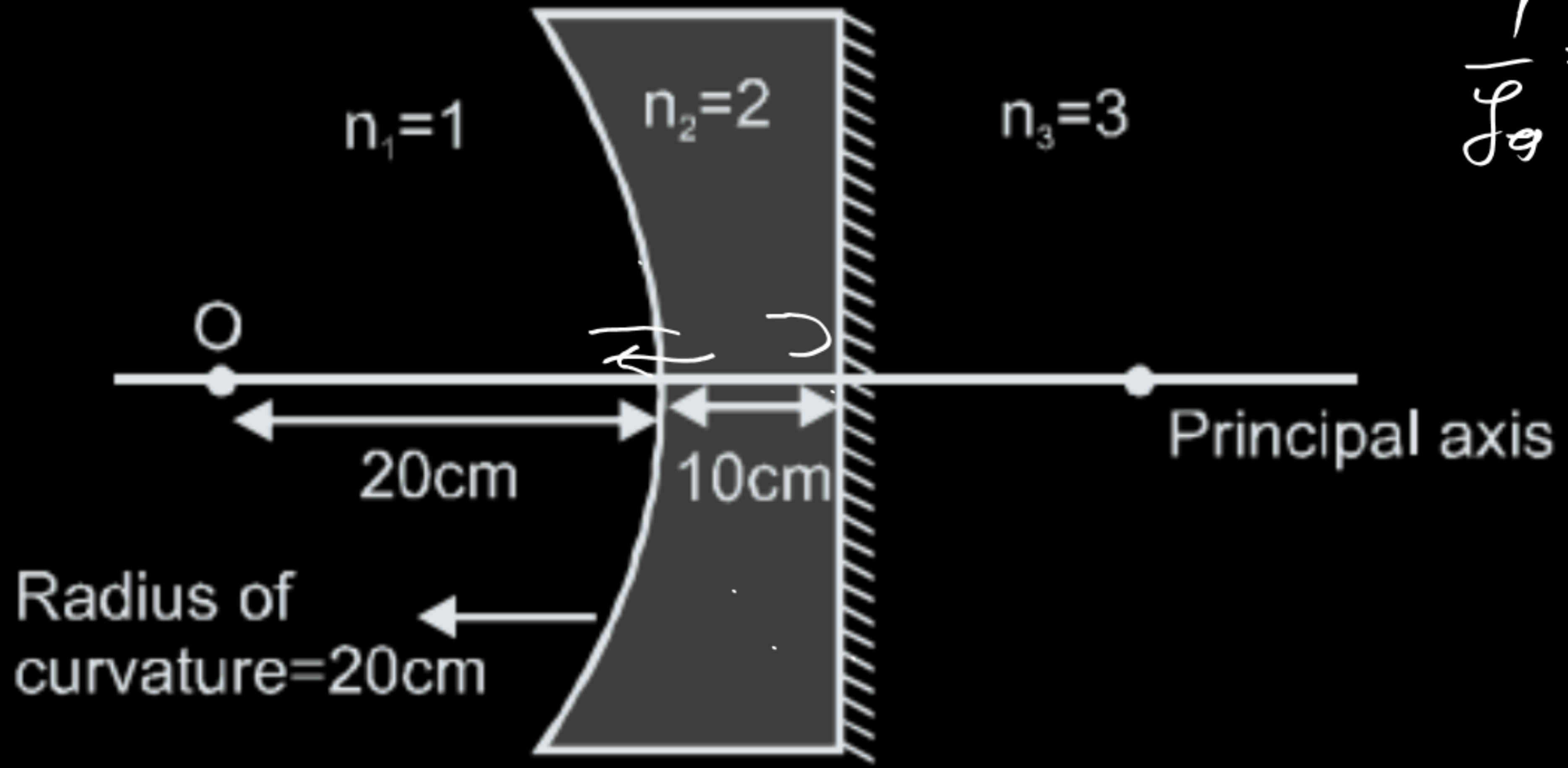
$$f = Em$$

$$= \frac{\cancel{2}\pi d G R}{3} \times \frac{\cancel{4}\pi}{3} \frac{R^3}{\cancel{8}_{2}} \frac{d}{\cancel{2}}$$

$$= \frac{\pi^2 R^4 d^2 G}{18}$$

Final image of point object 'O' formed by the combination is located :

$\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{2}{f_l}$



(A) On plane surface

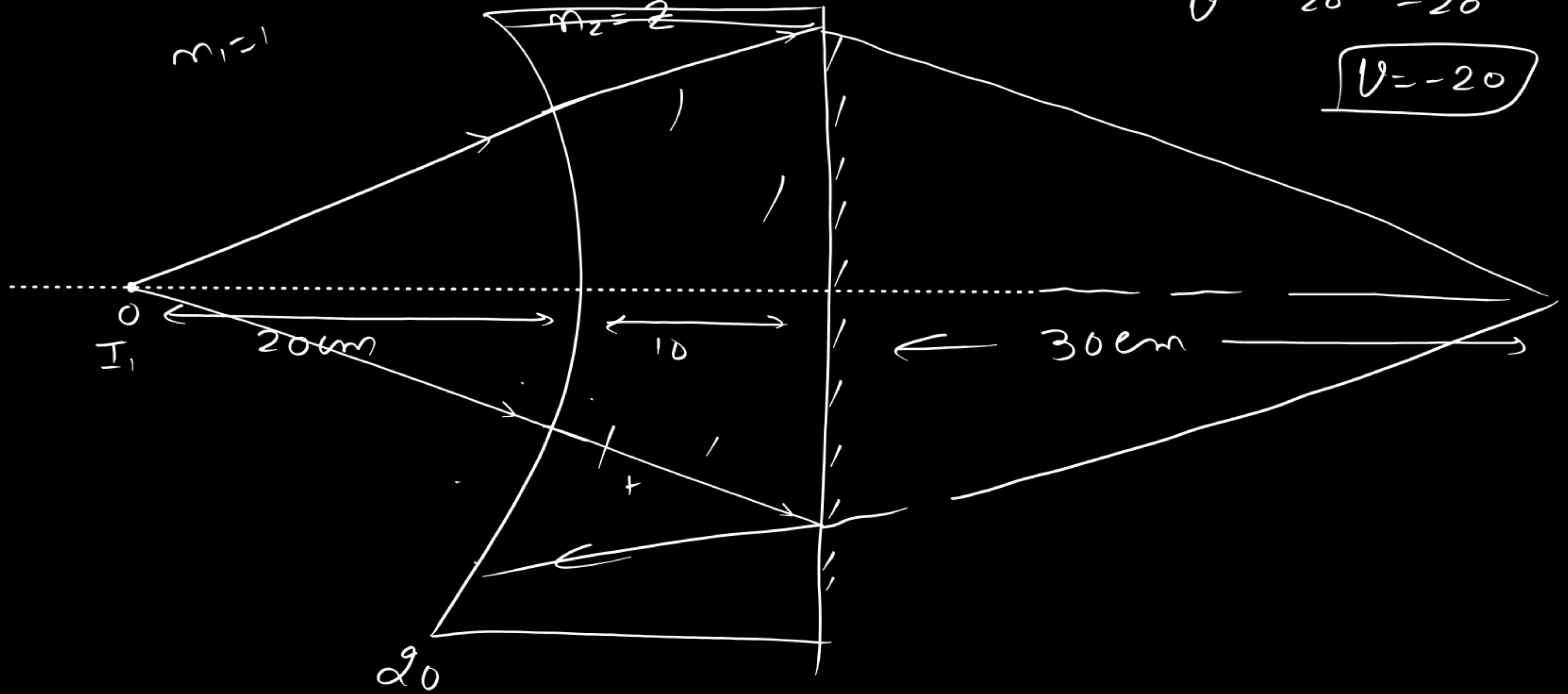
(B) at a distance 10cm from plane surface

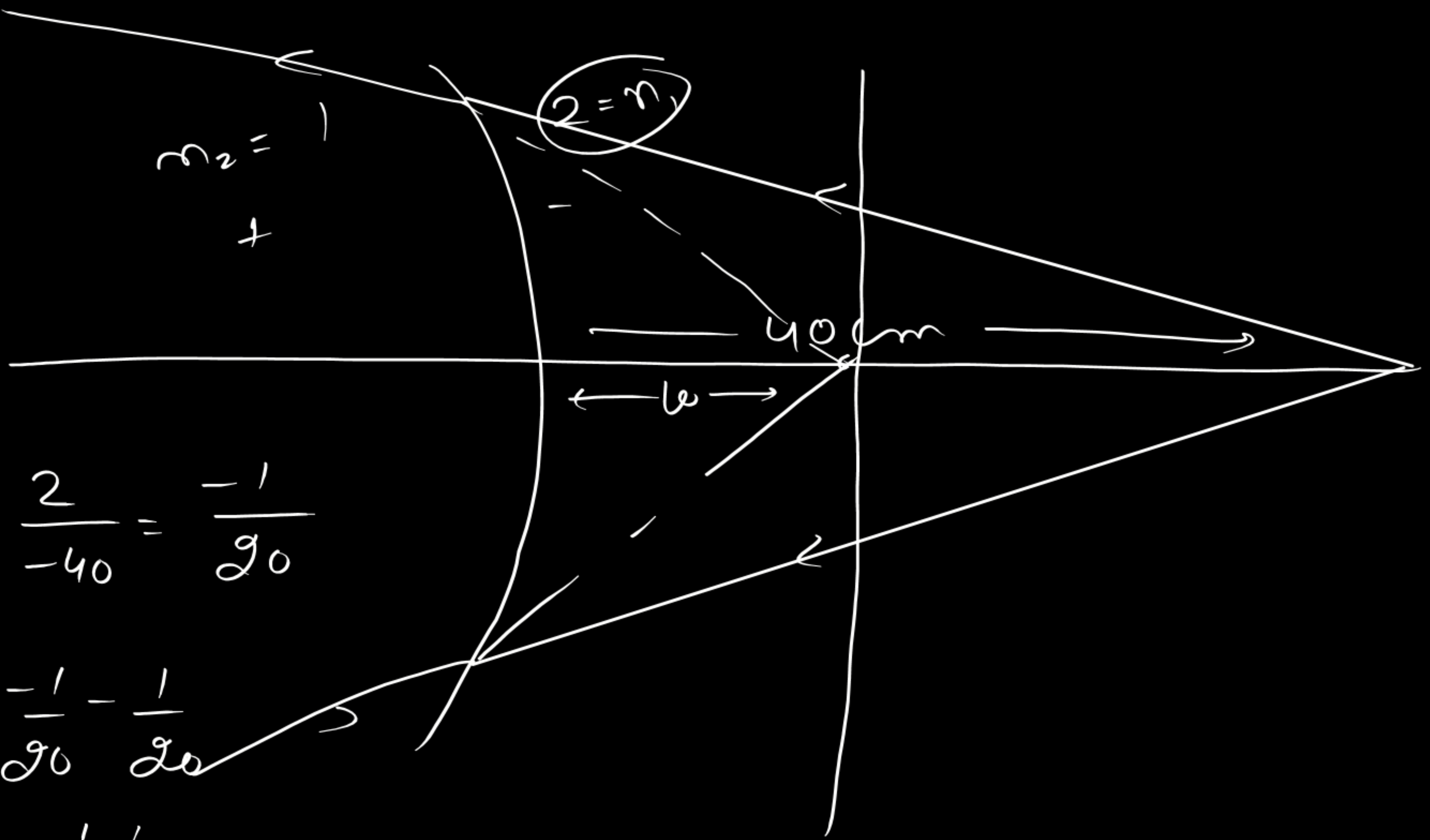
(C) at a distance 30cm from plane surface

(D) at a distance 20cm from curved surface

$$\frac{2}{v} - \frac{1}{-20} = \frac{1}{-20}$$

$$\boxed{v = -20}$$



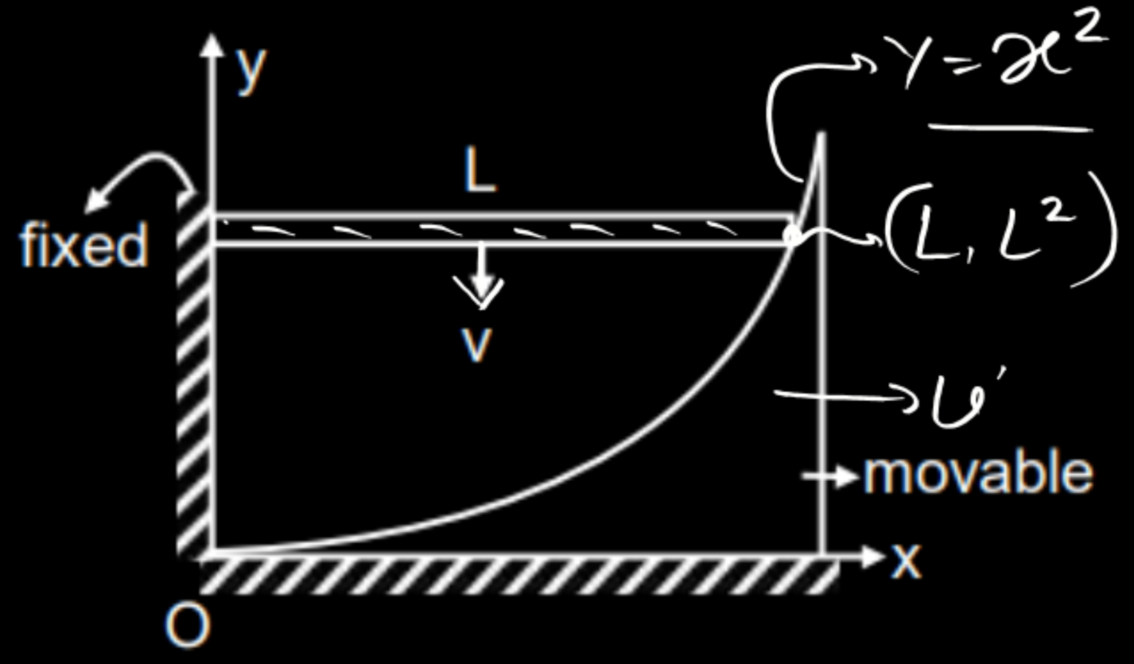


$$\frac{1}{v} - \frac{2}{-40} = \frac{-1}{20}$$

$$\frac{1}{v} = \frac{-1}{20} - \frac{1}{20}$$

$$= \frac{-1-1}{20} \Rightarrow v = -10$$

Consider a wedge rod system shown in the figure. Rod is moving vertically downwards without tilting with constant speed of v. The situation at t = 0 is shown in the diagram. The surface of the wedge is parabolic and has an equation $y = x^2$ at t = 0. Find the speed of wedge at t = 0 (Assume that the contact between the rod and the wedge is not lost)



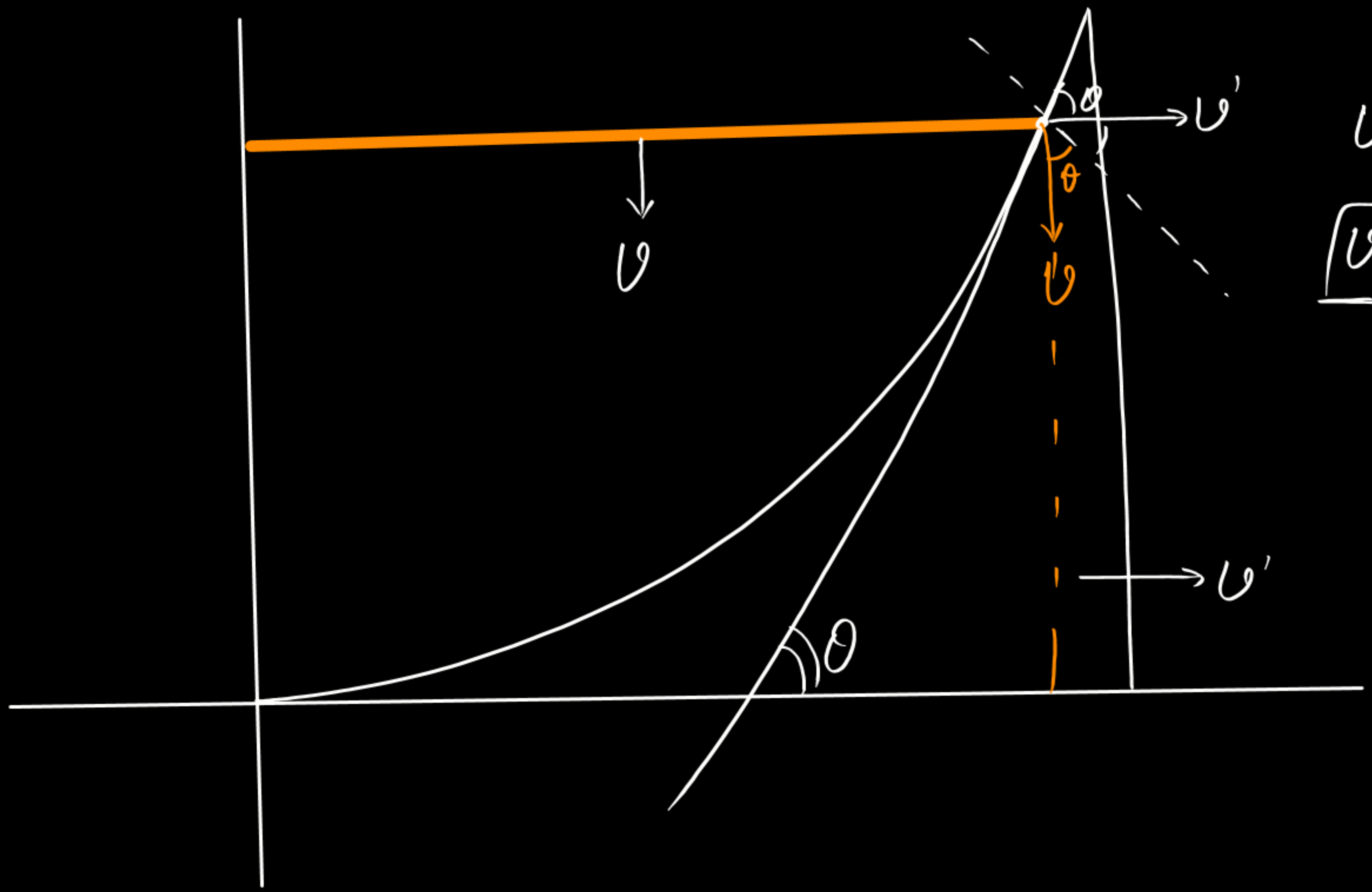
[at t = 0 wedge just touches the y-axis]

(A) 2vL

(B) $\frac{v}{2L}$

(C) $\frac{2v}{L}$

(D) vL



$$v' \sin\theta = v\cos\theta$$

$$\boxed{v' = v\cot\theta}$$

$$\tan\theta = \frac{dy}{dx} = \frac{d}{dx}(x^2)$$

$$= 2x$$

$$= 2L$$

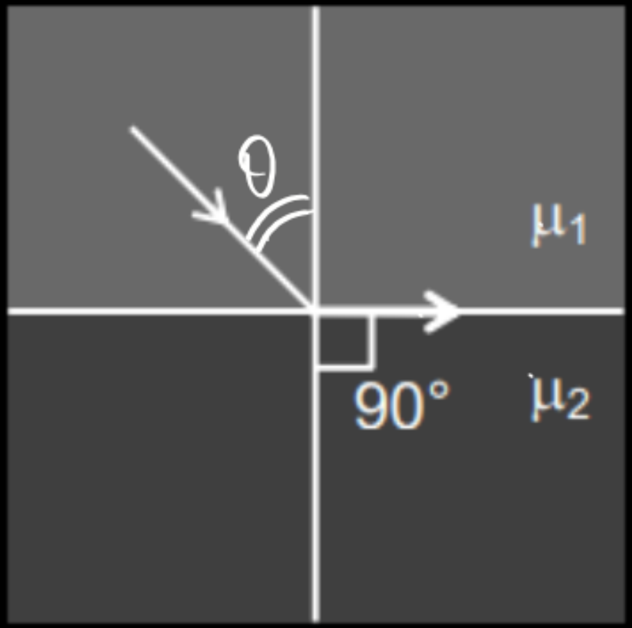
$$\boxed{v' = \frac{v}{2L}}$$

A ray is incident on interface of two media at critical angle as shown in the figure.

$\sin C = \frac{\mu_2 + x}{\mu_1 + x} > \frac{\mu_2}{\mu_1}$

$\sin c > \sin\theta$

$\boxed{c > \theta}$



$\mu_1 \sin\theta = \mu_2 \sin 90^\circ$

$\boxed{\sin\theta = \frac{\mu_2}{\mu_1}}$

$\sin c = \frac{2\mu_2}{2\mu_1} = \frac{\mu_2}{\mu_1}$

(A) If μ_1 and μ_2 both are increased by same amount the ray will suffer total internal reflection

(B) If μ_1 and μ_2 both are increased by same amount the ray will suffer refraction and angle of refraction will be less then 90°.

$\frac{1}{2}$: $\frac{2}{3}$ $\frac{3}{4}$

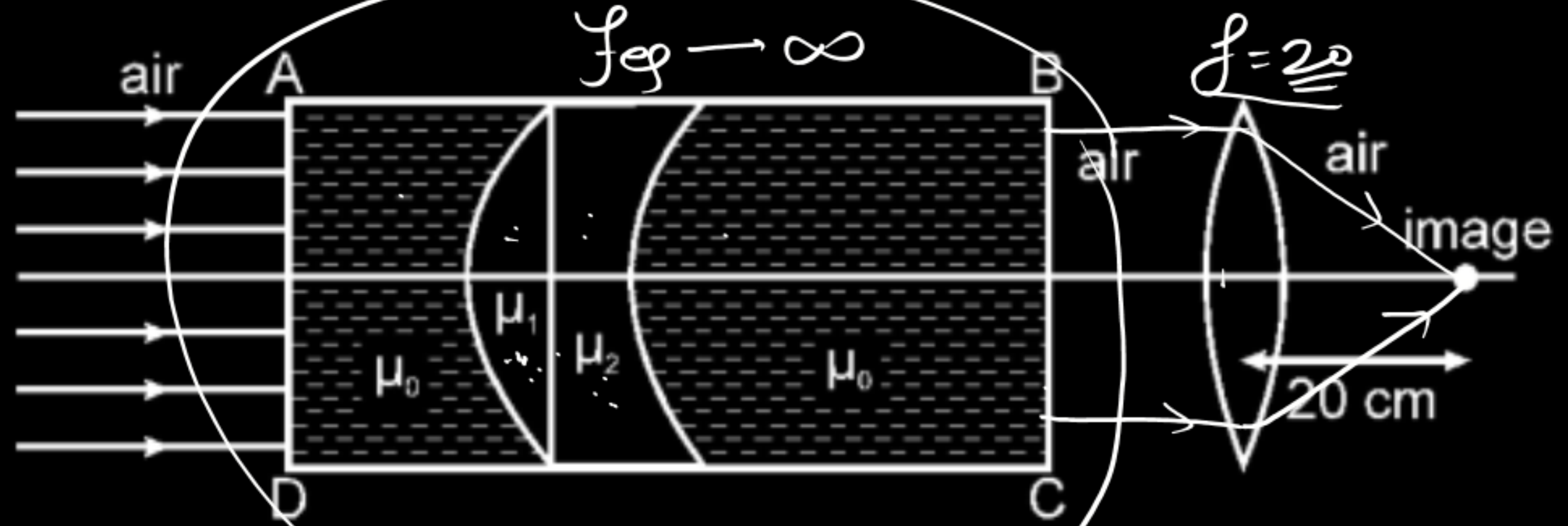
0.5 0.67 0.75

(C) If μ_1 and μ_2 both are doubled ray will suffer total internal reflection

(D) If μ_1 and μ_2 both are doubled ray will suffer refraction and angle of refraction will be less then 90°.

A parallel beam of monochromatic light is incident on face AD of the box ABCD, which contains a thin piano convex and a thin plano-concave lens of refractive indices μ_1 and μ_2 respectively. The lenses are coaxial and are submerged in a liquid of refractive index μ_0 as shown in the figure. The lenses have equal radius of curvature of their curved surfaces. The box is kept in air co–axially with a thin convex lens of focal length 20 cm. The final point image of the beam after passing through the box and the lens is formed 20 cm beyond the convex lens on the axis itself. If $\mu_1 = \mu_0 + \frac{b}{\lambda^2}$ and $\mu_2 = \mu_0 + \frac{c}{\lambda^4}$ where b and c are positive constants and λ is wavelength of light in air. Find the value of λ for which the above observation is possible.

$f_{eq} \to \infty \Rightarrow \frac{1}{f_{eq}} = 0 = \frac{1}{f_1} + \frac{1}{f_2}$



f

$\mu_1 = \mu_2$

$\frac{b}{\lambda^2} = \frac{c}{\lambda^4}$

$\lambda^2 = \frac{c}{b}$

$\boxed{\lambda = \sqrt{\frac{c}{b}}}$

$\left(\frac{\mu_1}{\mu_0} - 1\right)\left\{\frac{1}{R} - \frac{1}{\infty}\right\} + \left(\frac{\mu_2}{\mu_0} - 1\right)\left(\frac{1}{\infty} - \frac{1}{R}\right) = 0$

$\frac{\mu_1 - \mu_0}{\mu_0} \times \cancel{\frac{1}{R}} = \frac{\mu_2 - \mu_0}{\mu_0} \cancel{\frac{1}{R}}$

(A) $\frac{\sqrt{c}}{b}$ (B) $\sqrt{\frac{c}{b}}$ (C) $\frac{\sqrt{b}}{c}$ (D) $\sqrt{\frac{b}{c}}$

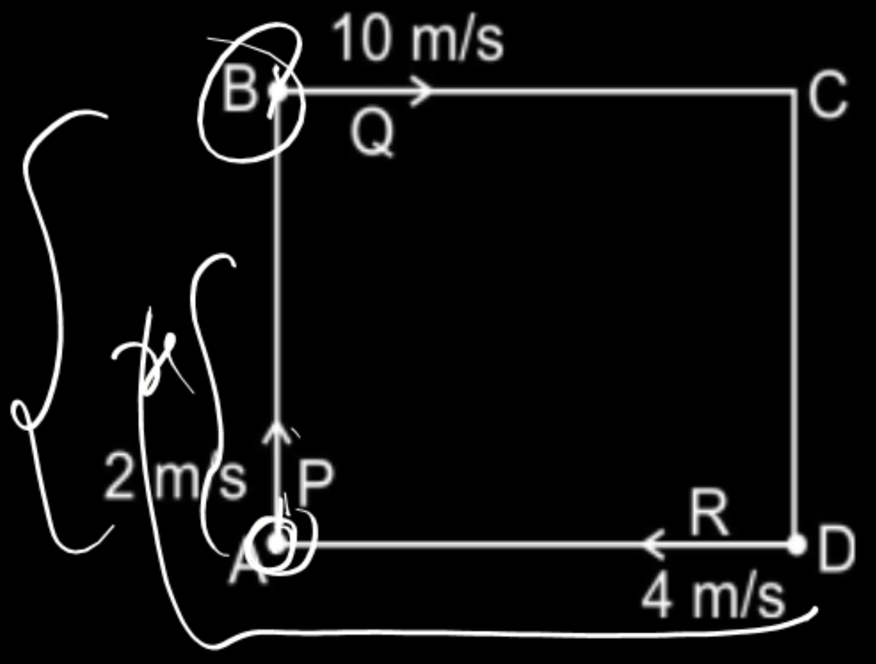
Three men P, Q & R are standing at corner A, B & D of square ABCD of side 8 m. They start moving along the track with constant speed 2 m/s, 10 m/s & 4 m/s respectively.

$x = 2t$

$24 + x = 10t$

$24 = 8t$

$t = 3$



(A) P & Q will meet 3 sec after start.

(B) P & R will meet 4 sec after start.

(C) P & R will meet at point B.

~~(D)~~ P & R will never meet.

$x = 2t$

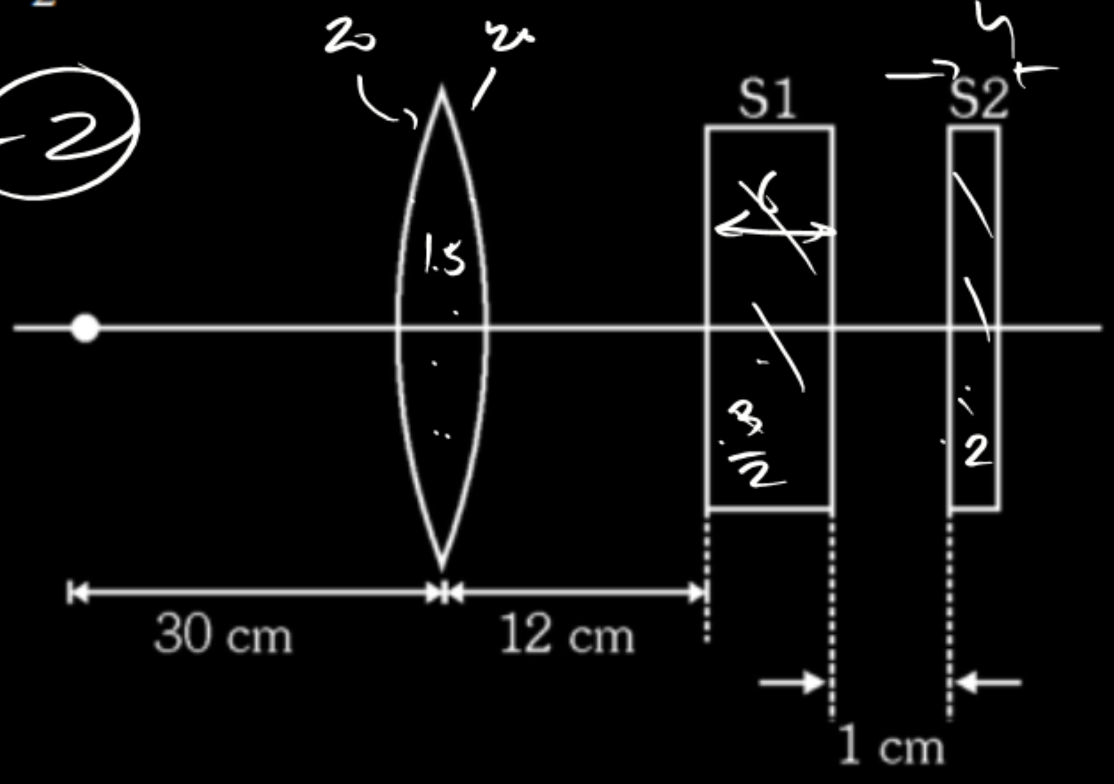
$8 + x = 4t$

$8 = 2t$

$\boxed{t = 4}$

A small object (A) is placed on the principal axis of an equiconvex lens at a distance of 30 cm. The refractive indes of the glass of the lens is 1.5 and its surfaces have radius of curvature R = 20 cm. Two glass slabs S_1 and S_2 have been place behind the lens as shown in figure. Thickness of the two slabs is 6 cm and 4 cm respectively and their refractive indices are $\frac{3}{2}$ and 2 respectively.

$$m = \frac{v}{u} = \frac{60}{-30} = -2$$



$$\frac{1}{f} = (1.5-1)\left\{\frac{1}{20} - \frac{1}{-20}\right\}$$

$$f = 20$$

$$\frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{3-2}{60}$$

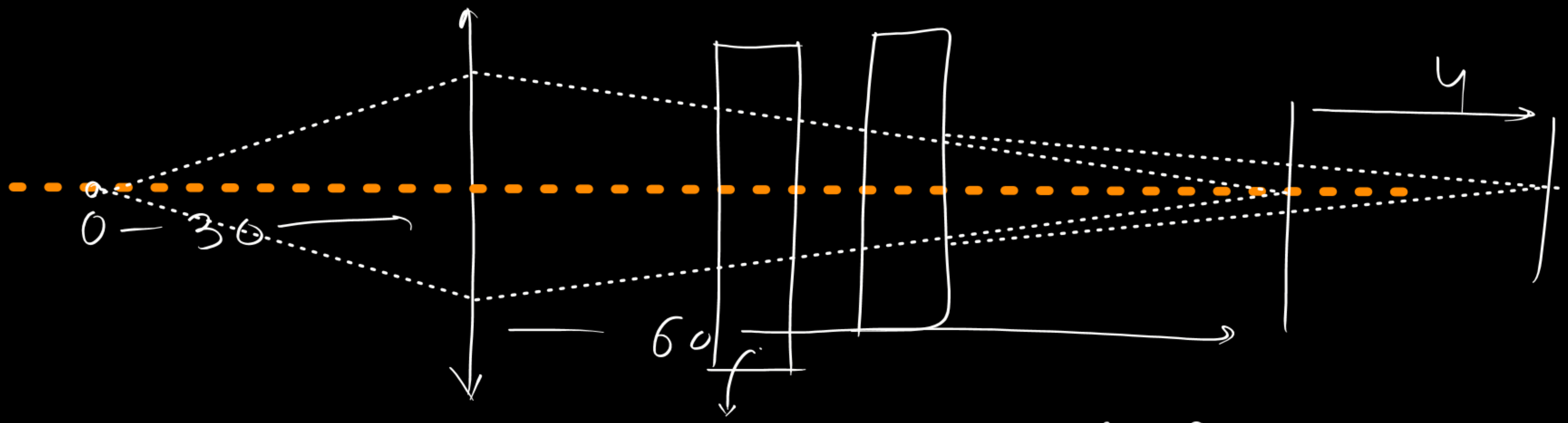
$$v = 60$$

(A) The distance of the final image measured from the lens is 64 cm

(B) The magnification is –2

(C) The distance of the final image measured from the lens is 32 cm

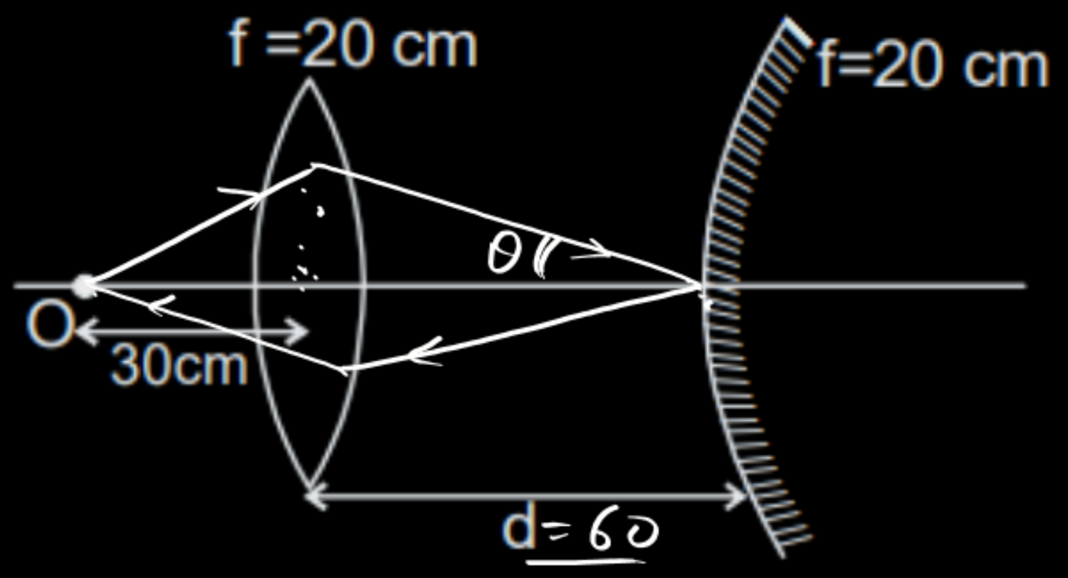
(D) The position of the image remain unchanged if the slab S_2 is moved to left so as to put it in contact with S_1



$$\text{shift} = t\left\{1 - \frac{1}{n}\right\} = 6\left\{1 - \frac{2}{3}\right\} = 2$$

$$= 4\left\{1 - \frac{1}{2}\right\} = 2$$

A convex lens and convex mirror are placed coaxially and seperated by a distance d. The focal length of both of them is 20 cm. A point object is placed at a distance 30 cm from the lens as shown in figure. Then the value of d so that image is formed on the object itself is



$$\frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{30}$$

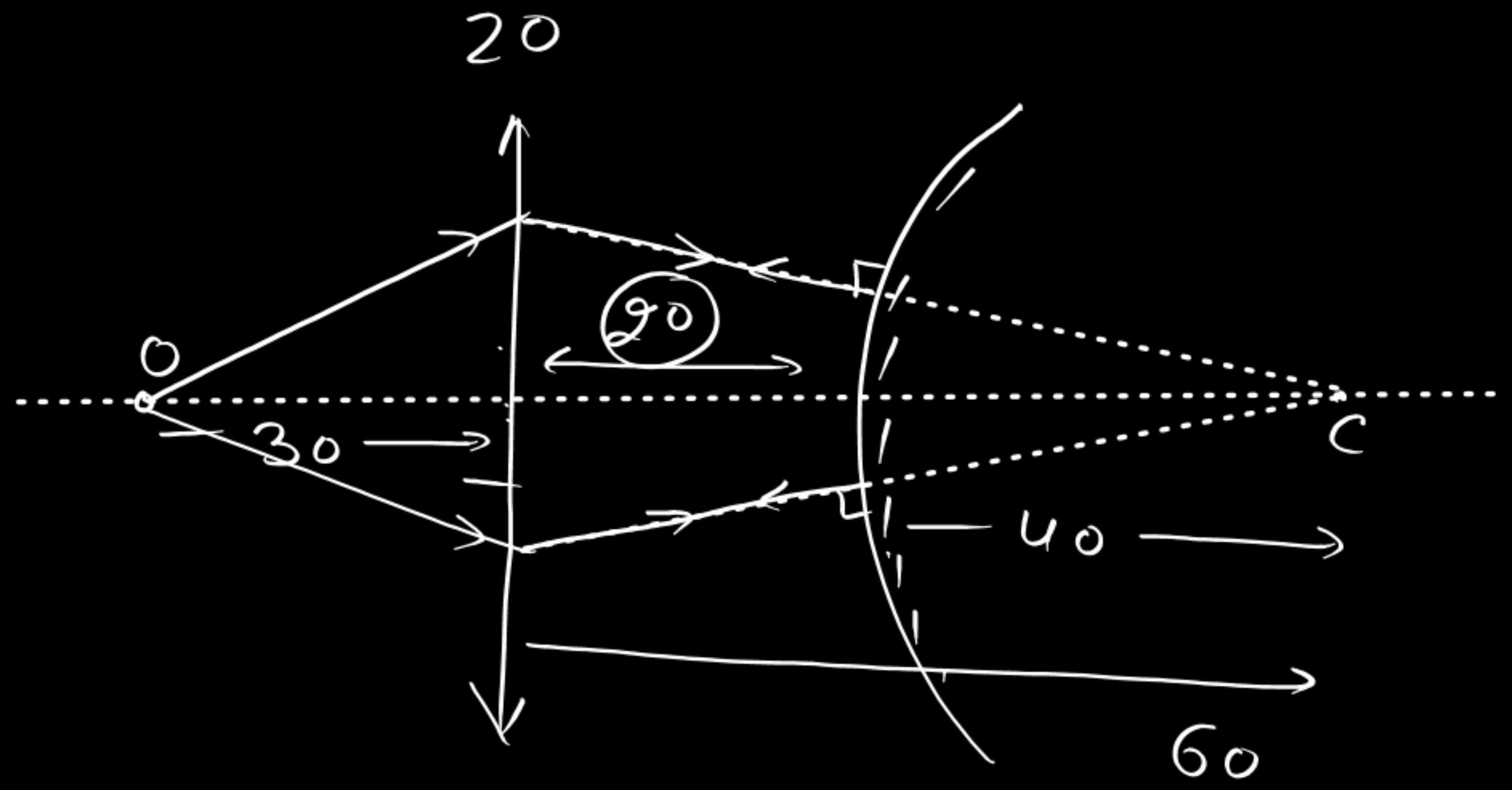
$$\boxed{v = +60}$$

(A) 10 cm

(B) 60cm

(C) 30 cm

(D) 20 cm


20
O
30
90
40
C
60

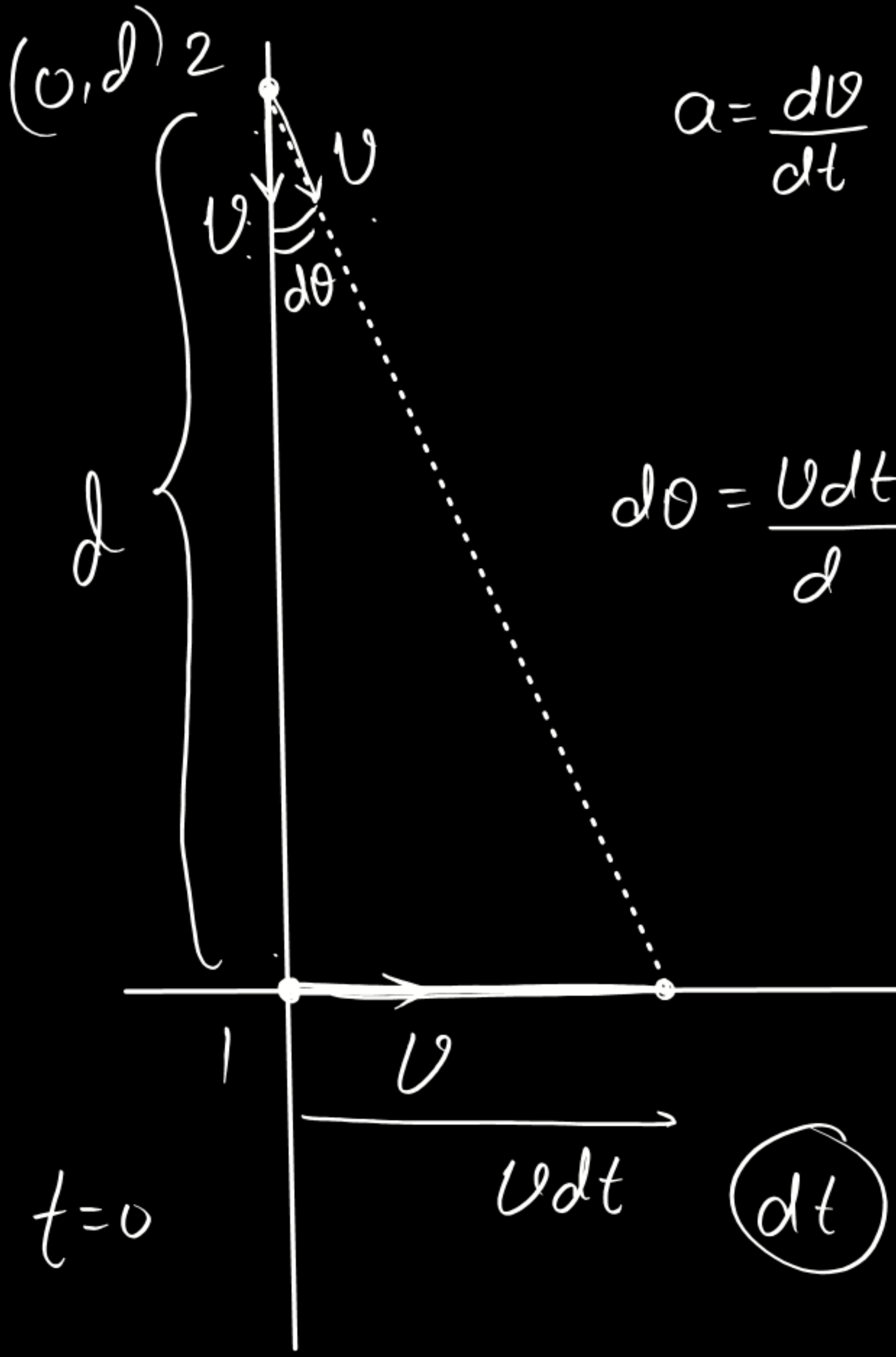
A particle P_1 moves with a constant velocity v along x-axis, starting from origin. Another particle P_2 chases particle P_1 with constant speed v, starting from the point (0, d). Both motion begin simultaneously. Select the correct alternative.

(A) Initial acceleration of P_2 is $\frac{v^2}{2d}$

(B) Ultimate separation between P_1 and P_2 is $\frac{d}{2}$

(C) Initial acceleration of P_2 is $\frac{v^2}{d}$

(D) Ultimate separation between P_1 and P_2 is $\frac{d}{3}$



$$a = \frac{dv}{dt} = \frac{v\,d\theta}{dt} = \frac{v \cdot v}{d}$$

$$\boxed{= \frac{v^2}{d} = a}$$

$$d\theta = \frac{v\,dt}{d}$$

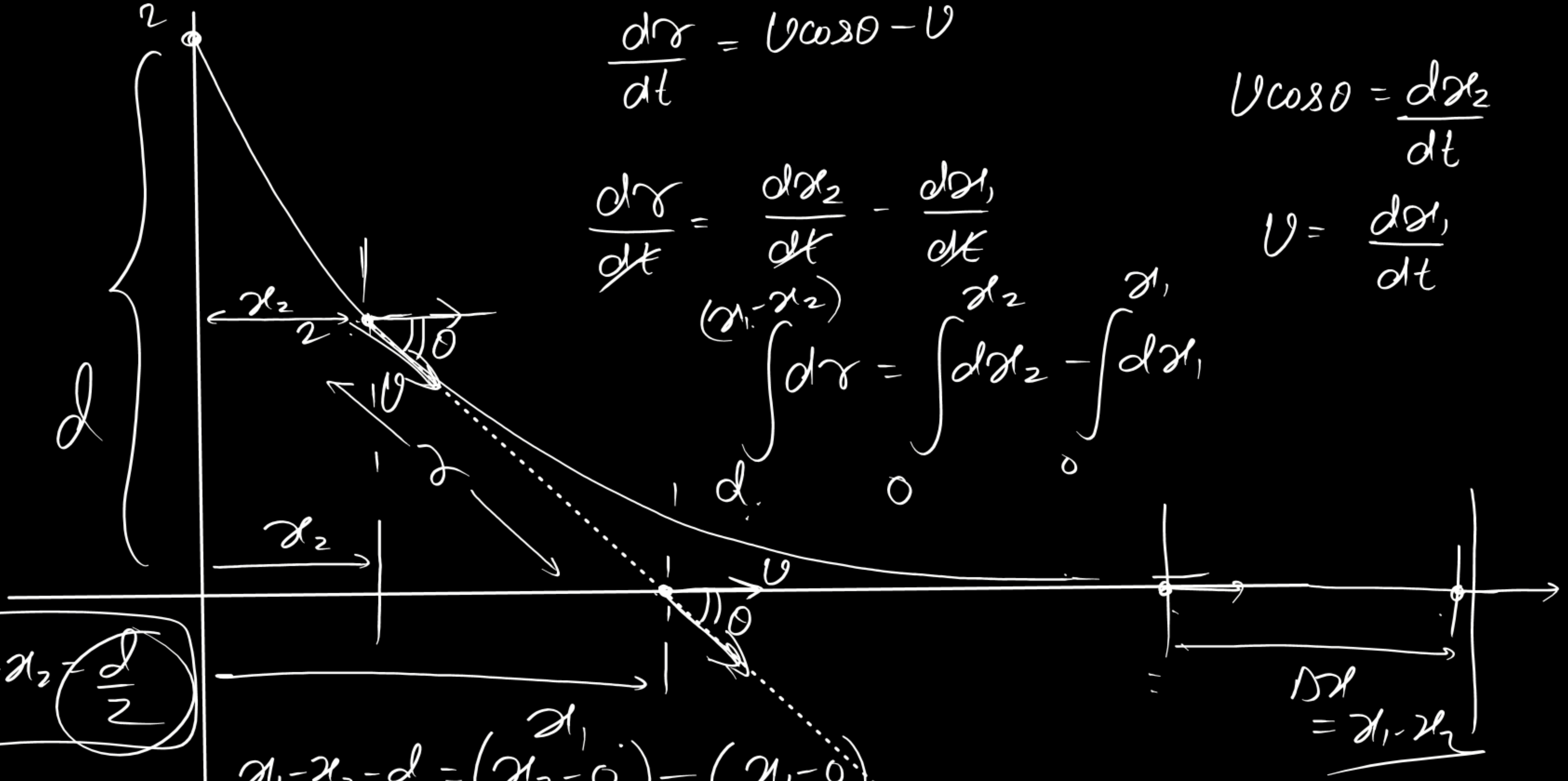
$$dv = |\vec{v}_f - \vec{v}_i|$$

$$= \sqrt{v^2 + v^2 - 2v^2 \cos d\theta}$$

$$= \sqrt{2v^2\{1 - \cos d\theta\}}$$

$$= \sqrt{2v^2 \times 2\sin^2\frac{d\theta}{2}}$$

$$= 2v \sin\frac{d\theta}{2} = \cancel{2} v \frac{d\theta}{\cancel{2}}$$



$$\frac{dr}{dt} = v\cos\theta - v$$

$$v\cos\theta = \frac{dx_2}{dt}$$

$$\frac{dr}{\cancel{dt}} = \frac{dx_2}{\cancel{dt}} - \frac{dx_1}{\cancel{dt}}$$

$$v = \frac{dx_1}{dt}$$

$$\int_d^{(x_1-x_2)} dr = \int_0^{x_2} dx_2 - \int_0^{x_1} dx_1$$

$$x_1 - x_2 - d = (x_2 - 0) - (x_1 - 0)$$

$$2(x_1 - x_2) = d$$

$$\boxed{x_1 - x_2 = \frac{d}{2}}$$

A stone is projected from level ground with initial speed u and at an angle θ with the horizontal. Somehow the acceleration due to gravity become four times immediately after the stone reaches the maximum height and remains same there after. (Acceleration due to gravity is 'g')

$$\frac{u\sin\theta}{g} + \frac{u\sin\theta}{2g} = \frac{3u\sin\theta}{2g} \qquad t_a = \frac{u\sin\theta}{g}$$

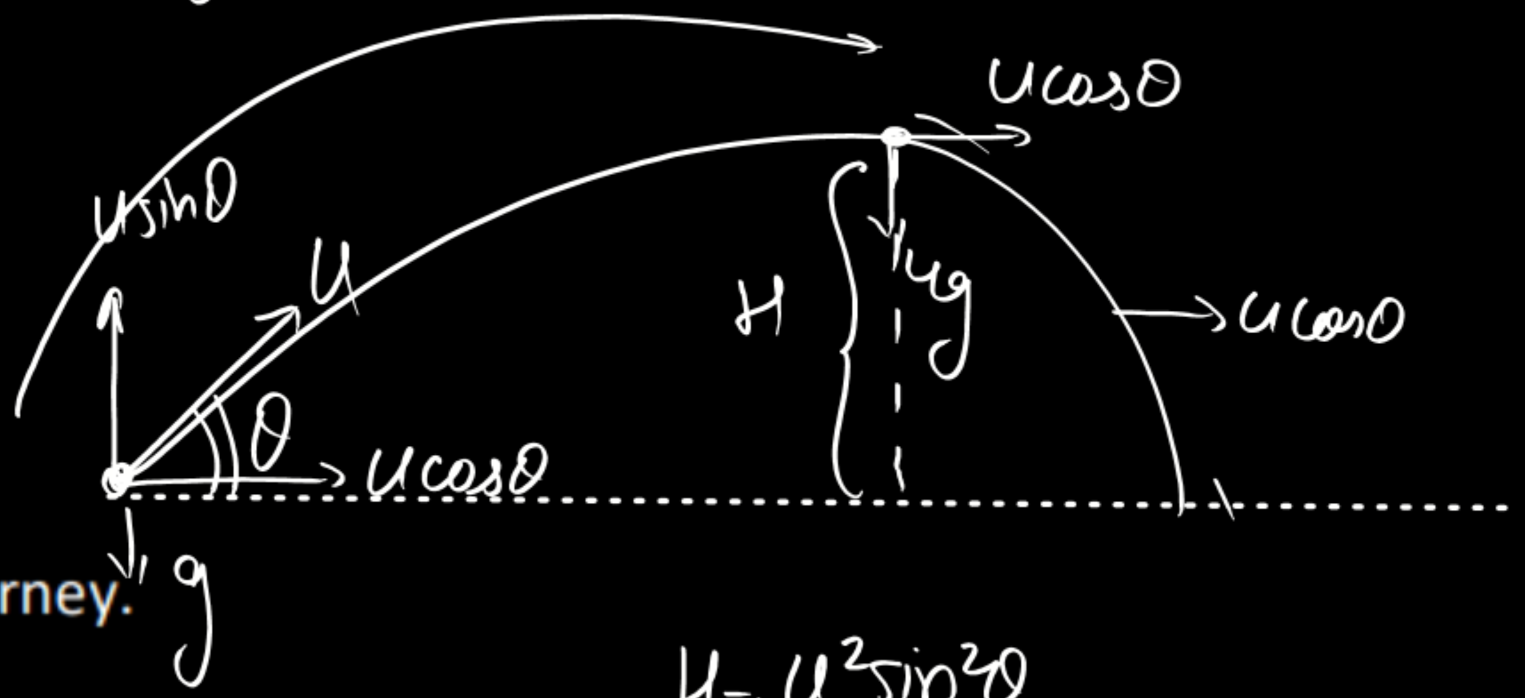
Choose the correct option :

(A) The total time of flight of particle is $\frac{3}{2}\frac{u\sin\theta}{g}$.

(B) The total time of flight of particle is $\frac{7}{4}\frac{u\sin\theta}{g}$.

(C) Time of upward Journey is equal to downward Journey.

(D) Time of upward Journey is less than downward Journey.



$$H = \frac{u^2\sin^2\theta}{2g}$$

$$t_d = \sqrt{\frac{2H}{g'}} = \sqrt{\frac{\cancel{2}\cdot u^2\sin^2\theta}{\cancel{2}g \times 4g}} = \frac{u\sin\theta}{2g}$$

A stone is projected from level ground with initial speed u and at an angle θ with the horizontal. Somehow the acceleration due to gravity become four times immediately after the stone reaches the maximum height and remains same there after. (Acceleration due to gravity is 'g')

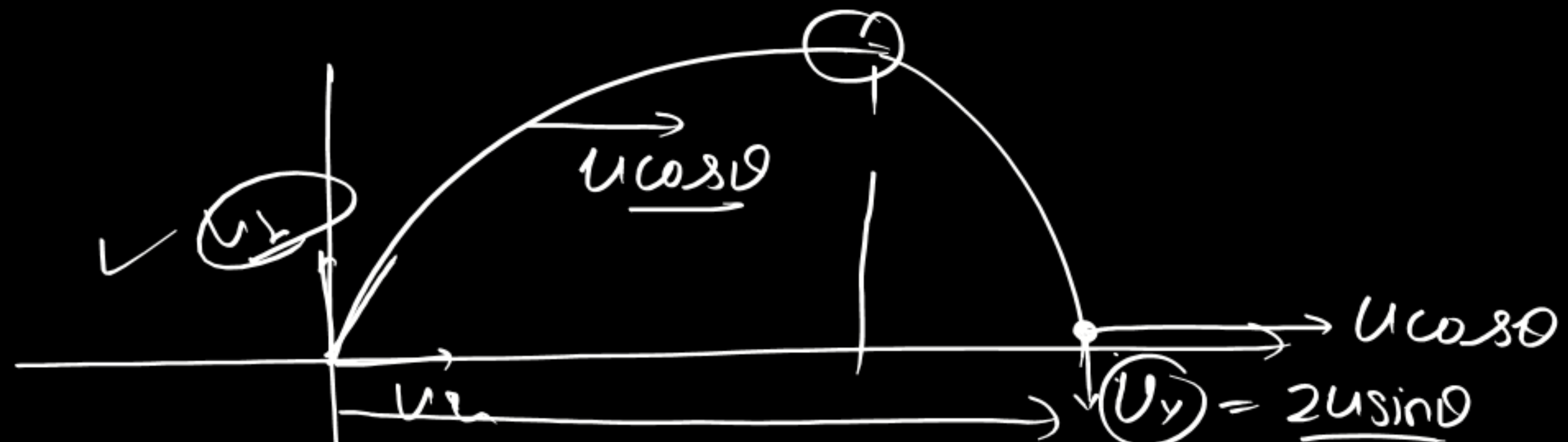
Choose the incorrect option :

(A) The horizontal range of particle is $\frac{3}{4}\frac{u^2\sin 2\theta}{g}$.

(B) The horizontal range of particle is $\frac{7}{8}\frac{u^2\sin 2\theta}{g}$.

(C) Horizontal range is maximum at θ = 45°.

(D) Speed of particle just before it hits the ground is more than speed of projection.



$$R = u\cos\theta \times \frac{3u\sin\theta}{2g}$$

$$= \frac{3}{4}\frac{u^2\sin 2\theta}{g}$$

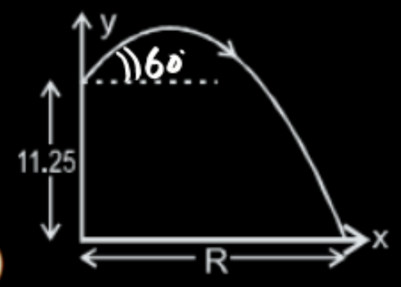
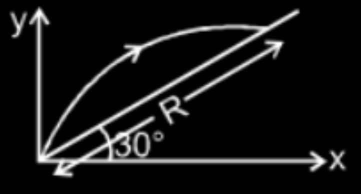
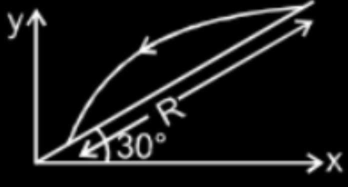
$$U_y = 0 + 4g \times \frac{u\sin\theta}{2g} = 2u\sin\theta$$

$$\sqrt{u^2\cos^2\theta + 4u\sin^2\theta}$$

$$= \sqrt{u^2 + 3u^2\sin^2\theta} > u$$

Match the Listing :

In the list-I, the path of a projectile (initial velocity 10 m/s and angle of projection with horizontal 60° in all cases) is shown in different cases. Rangle 'R' is to be matched in each case from list-II. Take g = 10 m/s². Arrow on the trajectory indicates the direction of motion of projectile.

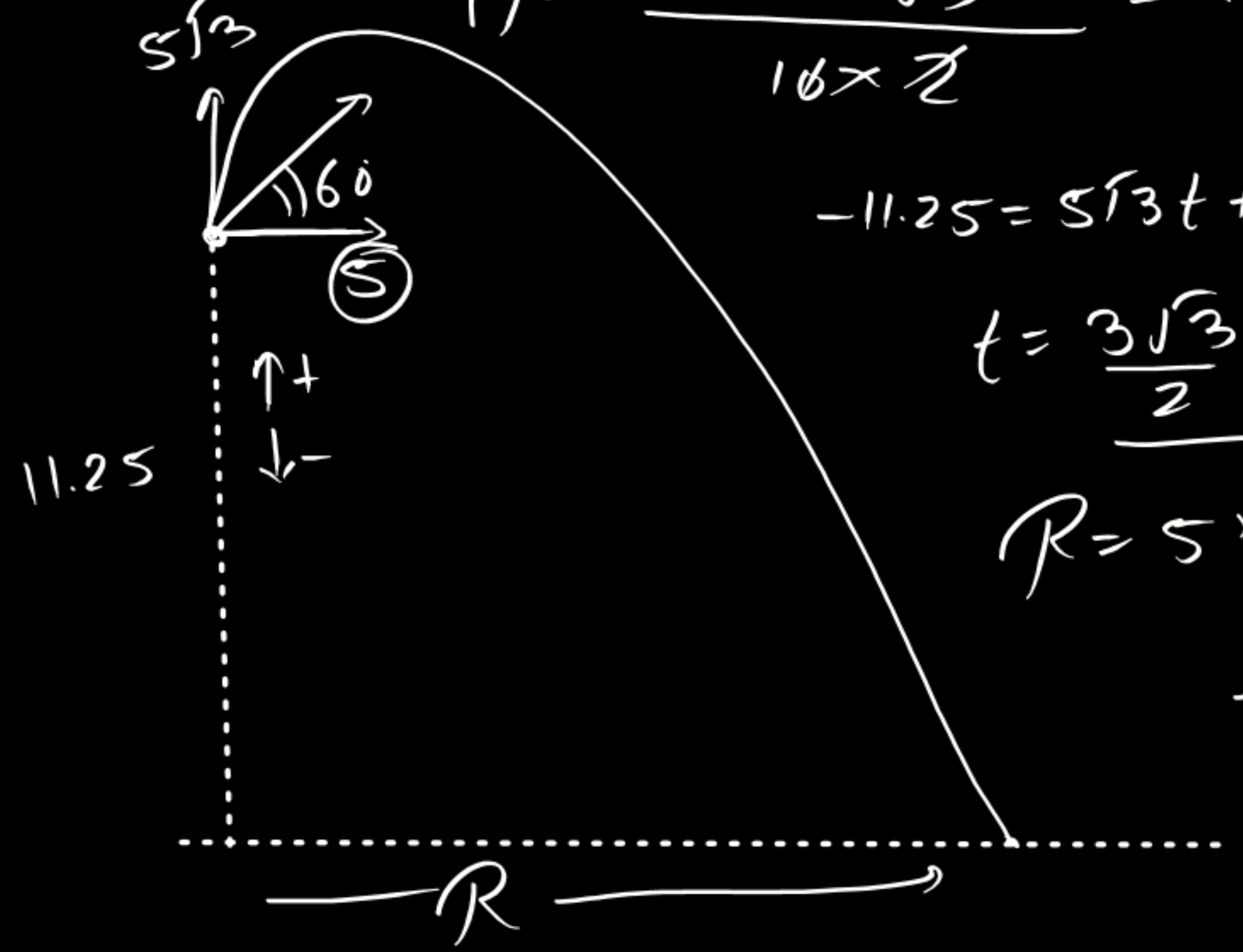
List-I	List-II
(P) 	(1) $R = \frac{15\sqrt{3}}{2}$ m
(Q) 	(2) $R = \frac{40}{3}$ m
(R) 	(3) $R = 5\sqrt{3}$ m
(S) 	(4) $R = \frac{20}{3}$ m

Codes :

	P	Q	R	S
~~(A)~~	~~4~~	~~2~~	~~3~~	~~1~~
~~(B)~~	~~3~~	~~2~~	~~1~~	~~4~~
(C)	3	1	4	2
(D)	3	1	2	4

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$R = \frac{\cancel{100}^{5} \times \sqrt{3}}{\cancel{10} \times \cancel{2}} = 5\sqrt{3}$$

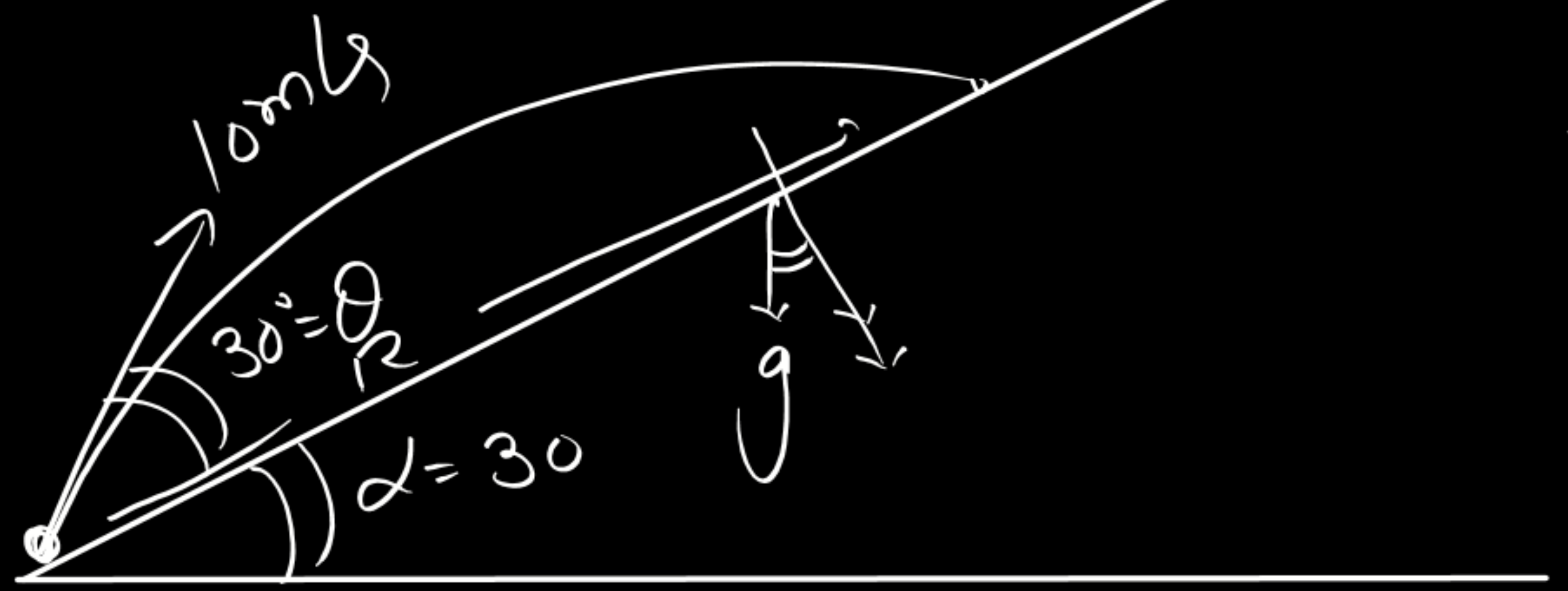


$$-11.25 = 5\sqrt{3}\,t + \frac{1}{2}(-10)t^2$$

$$t = \frac{3\sqrt{3}}{2} \text{ sec}$$

$$R = 5 \times \frac{3\sqrt{3}}{2}$$

$$= \frac{15\sqrt{3}}{2}$$



$$T = \frac{2u_1}{g_1} = \frac{2\times 8\times 2}{10\times\sqrt{3}} = \frac{2}{\sqrt{3}}$$

$$R = 5\sqrt{3}\times\frac{2}{\sqrt{3}} - \frac{1}{2}\times 5\times\frac{4}{3}\quad 2$$

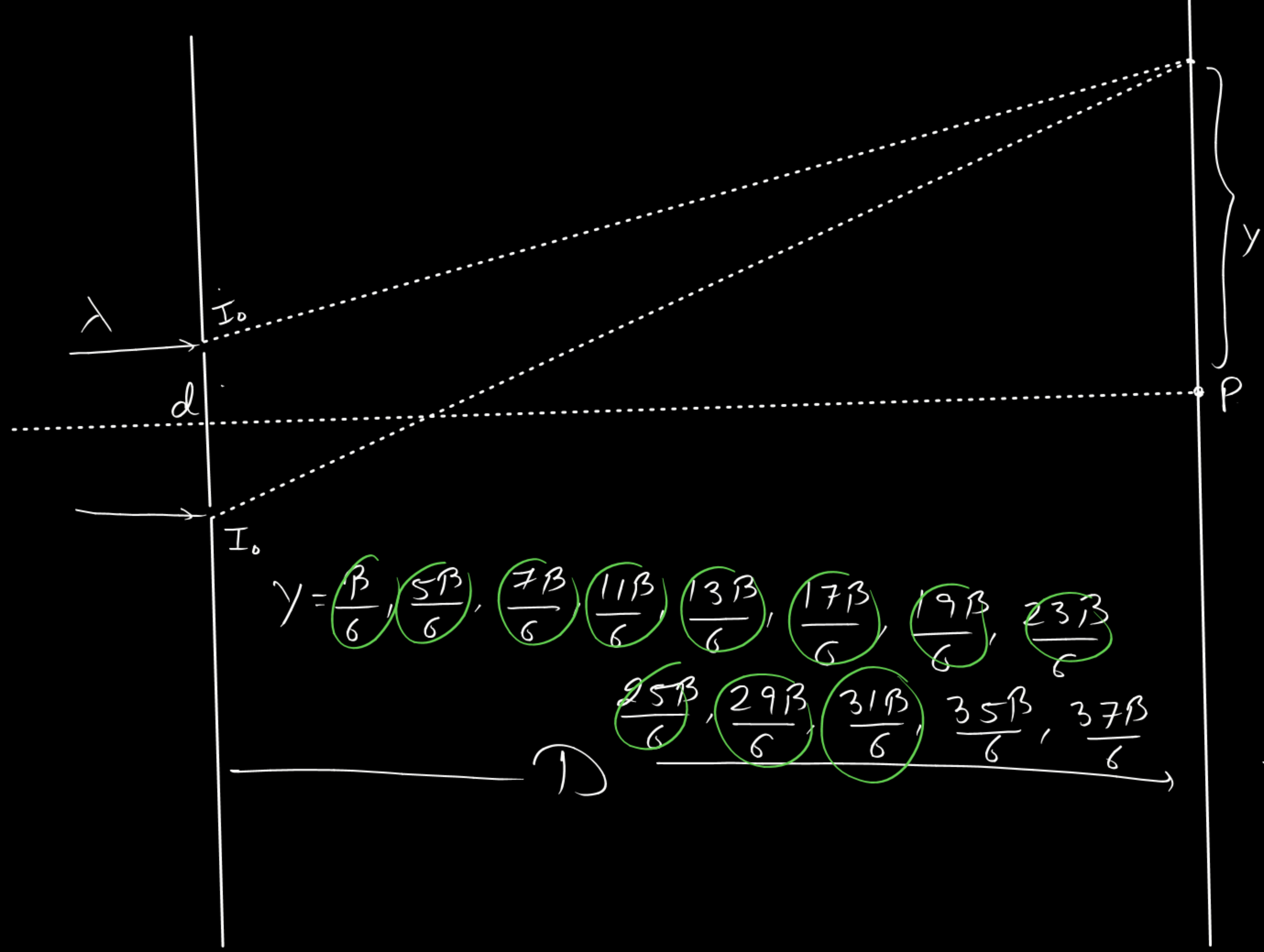
$$= 10 - \frac{10}{3} = \frac{20}{3}$$

In a Young's experiment, the upper slit is covered by a thin glass plate of refractive index $\mu = 1.4$ while the lower slit is covered by another glass plate having the same thickness as the first one but having refractive index $\mu_2 = 1.7$. Interference pattern is observed using light of wavelength $\lambda = 5400$ A. It is found that the point P on the screen where the central maxima fell before the glass plate were inserted, now 3/4 the original intensity, it is further observed that what used to be the fifth maxima earlier, lies below the point P while the sixth minima lies above the point P. Neglecting absorption of light by glass plates, calculate thickness of the glass plates in μm.

93×10^{-7} m

9.3×10^{-6} m

$9.3\ \mu m$



$$y = \frac{\beta}{6}, \frac{5\beta}{6}, \frac{7\beta}{6}, \frac{11\beta}{6}, \frac{13\beta}{6}, \frac{17\beta}{6}, \frac{19\beta}{6}, \frac{23\beta}{6}$$

$$\frac{25\beta}{6}, \frac{29\beta}{6}, \frac{31\beta}{6}, \frac{35\beta}{6}, \frac{37\beta}{6}$$

$$I_{Res} = I_0 + I_0 + 2I_0 \cos\Delta\phi$$

$$= 4I_0 \cos^2\frac{\Delta\phi}{2}$$

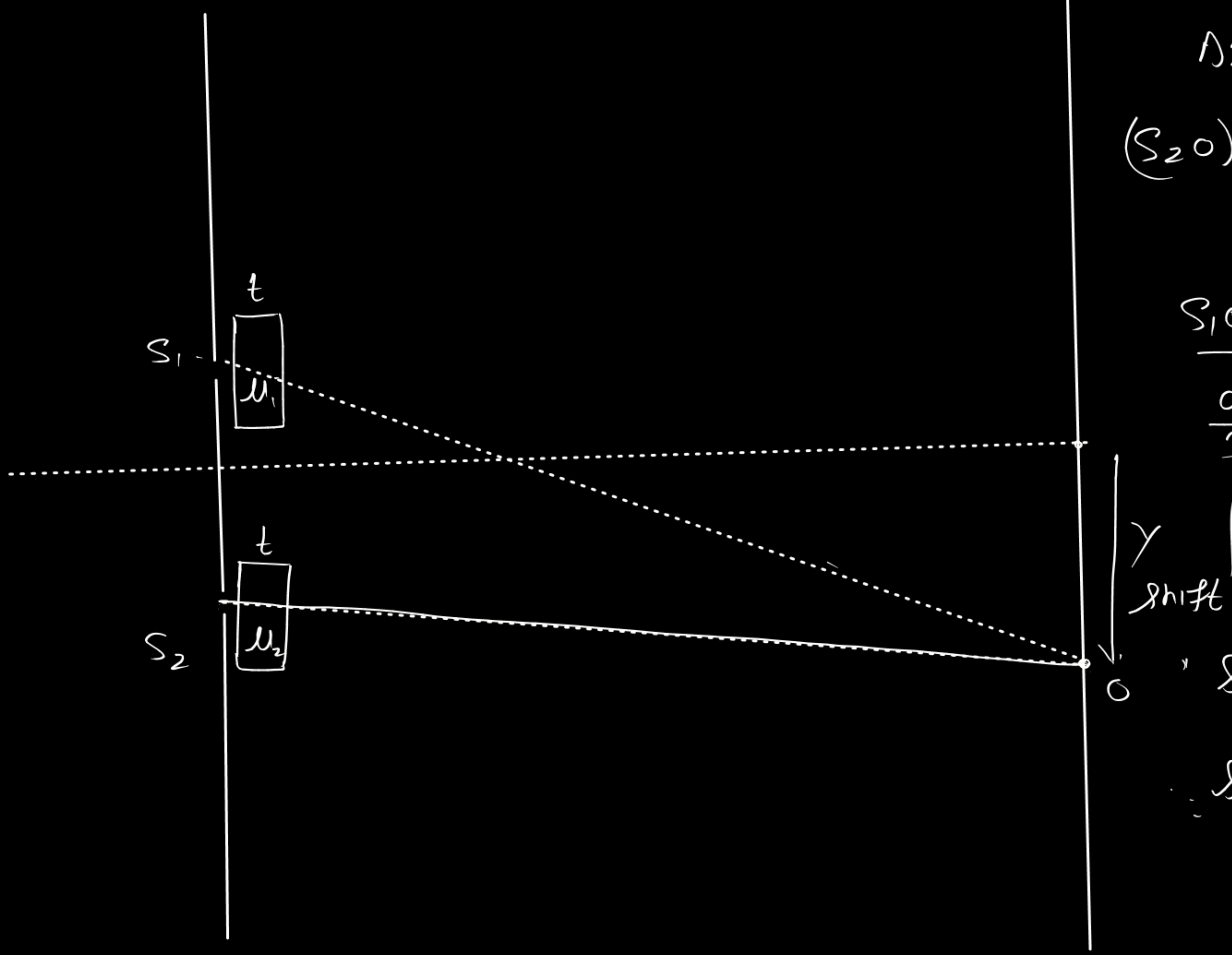
$$= 4I_0 \frac{2\pi\, d y}{2 \cdot \lambda D}$$

$$\boxed{I = 4I_0 \frac{\pi y}{\beta}}$$

$$3I_0 = 4I_0 \frac{\pi y}{\beta} \cos\frac{\pi y}{\beta}$$

$$\pm\frac{\sqrt{3}}{2} = \cos\frac{\pi y}{\beta} \quad \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6} \ldots$$



$$\Delta x = 0 = (S_2O)' - (S_1O)'$$

$$(S_2O) - \cancel{t} + \mu_2 t$$

$$= (S_1O) - \cancel{t} + \mu_1 t$$

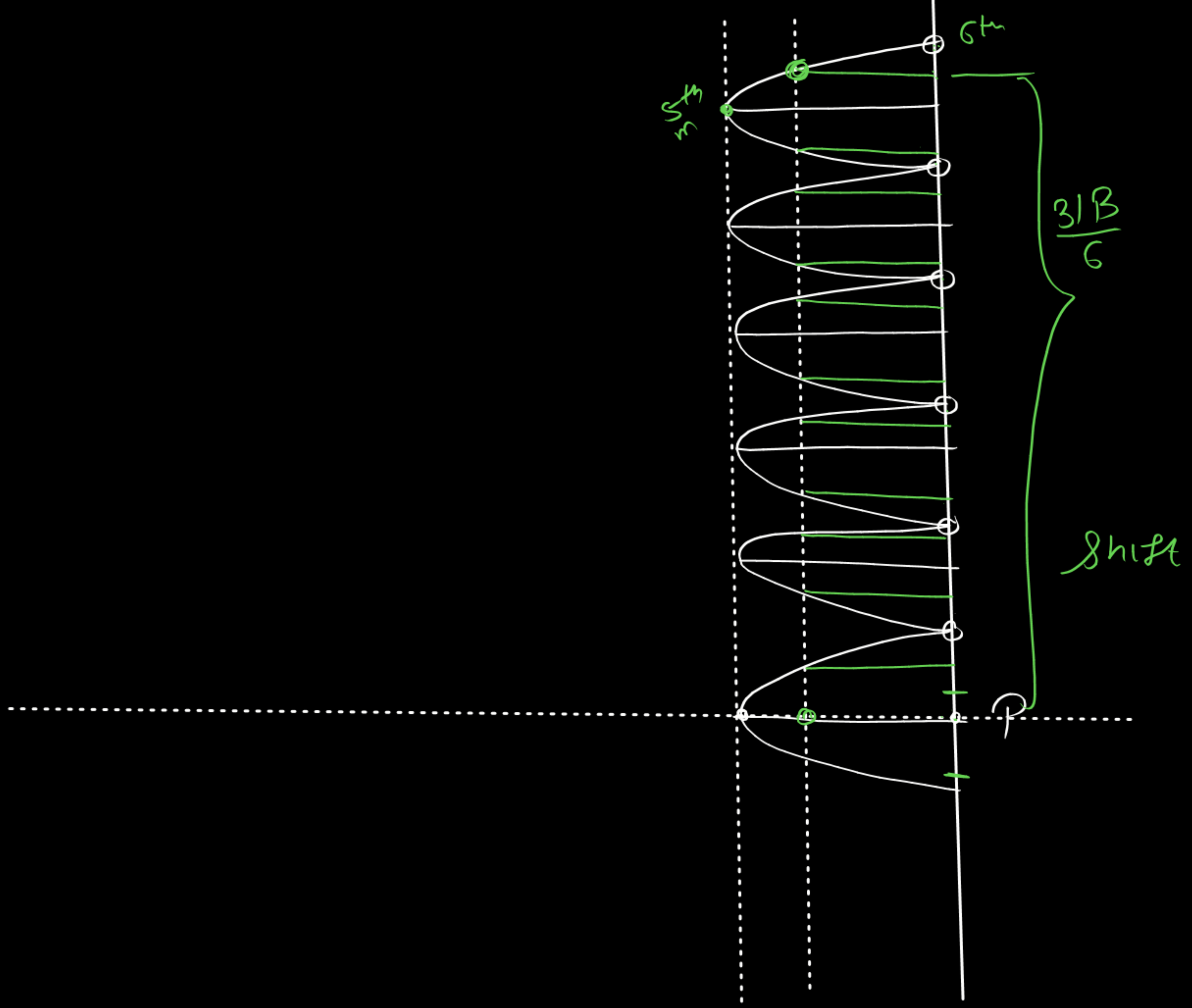
$$\underline{S_1O - S_2O} = (\mu_2 - \mu_1)t$$

$$\frac{dY}{D} = (\mu_2 - \mu_1)t$$

$$\boxed{Y = \frac{Dt(\mu_2 - \mu_1)}{d}}$$

$$\delta_1 = \frac{Dt(\mu_1 - 1)}{d}$$

$$\delta_2 = \frac{Dt(\mu_2 - 1)}{d}$$



$$\frac{31\lambda \cancel{D}}{6\,\cancel{d}} = \frac{(\mu_2-\mu_1)\cancel{D}t}{\cancel{d}}$$

$$t = \frac{31\lambda}{6(\mu_2-\mu_1)}$$

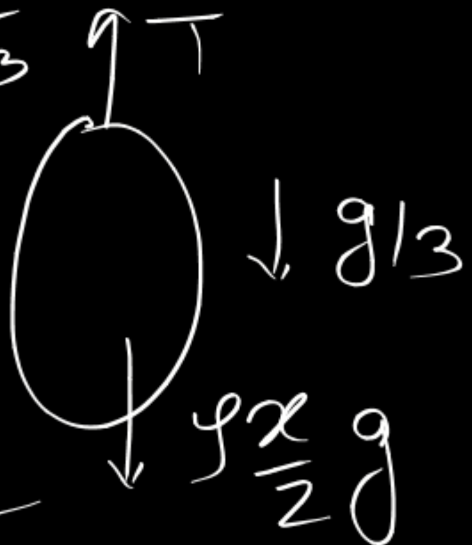
$$= \frac{31\times 5\cancel{4000}^{3000}\times 10^{-10}}{\cancel{6}\times 0.\cancel{3}}$$

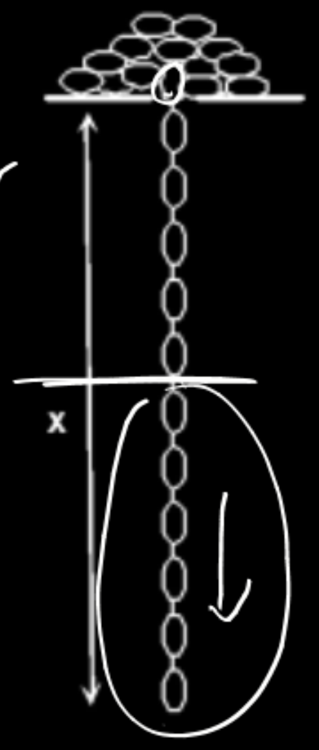
$$= 93\times 10^{-87}\,m$$

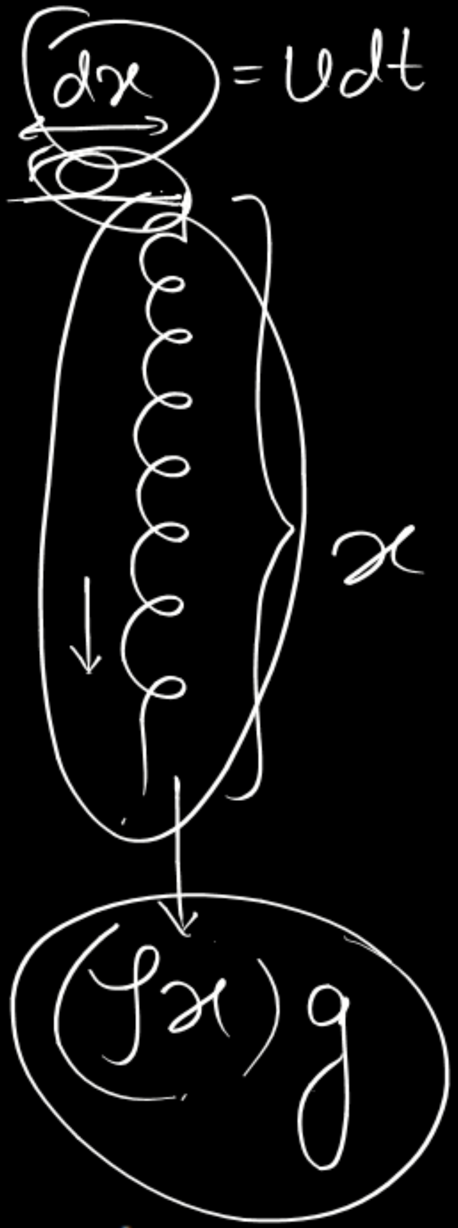
One end of the chain falls through a hole in its support and pulls the remaining links after it in a steady flow. If the links which are initially at rest, acquire the velocity of the chain suddenly and without frictional resistance or interference from the support or from adjacent links. Choose the **CORRECT**statement, (when x = 0, then v = 0) (length of the chain is L and ρ is the mass per unit length of the chain).

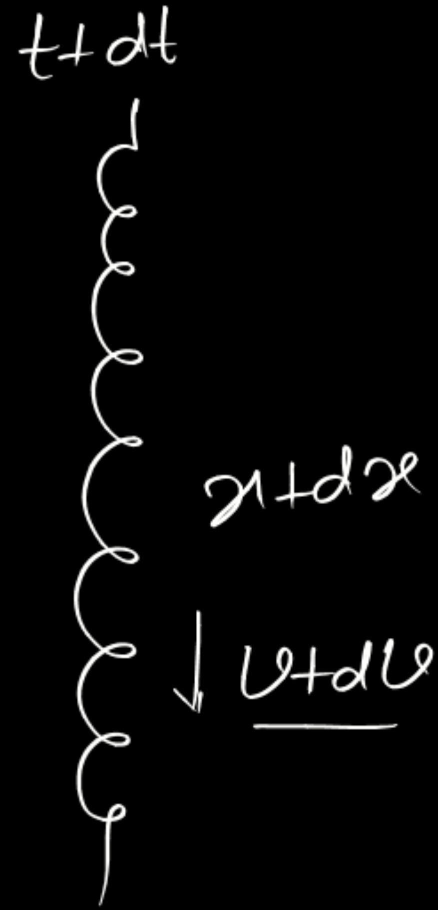
$$\rho \frac{x}{2} g - T = \rho \frac{x}{2} \cdot g/3$$

$$T = \frac{\rho g x}{\cancel{2}} \times \frac{\cancel{2}}{3}$$









(A) The velocity v of the chain as a function of x is $\sqrt{\frac{2gx}{3}}$

(B)The acceleration of a of the falling chain as a function of x is $\frac{g}{3}$

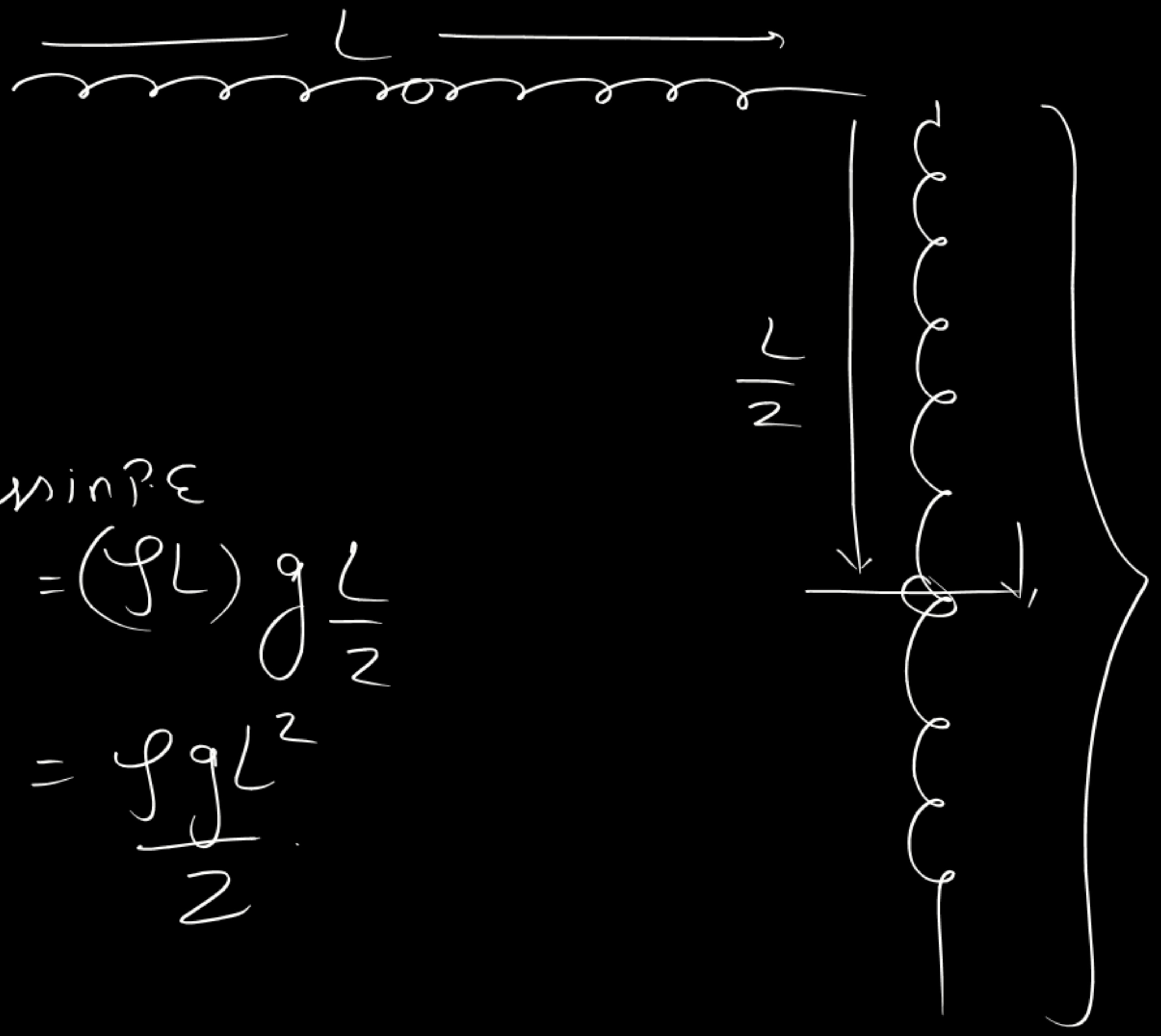
(C)The energy Q lost from the system as the last link leaves the platform is $\frac{\rho g L^2}{6}$

(D) Tension at the middle point of falling chain is $\frac{\rho g x}{3}$

$$v^2 = \frac{2}{3} g x$$

$$2v\frac{dv}{dx} = \frac{2}{3} g \quad (1)$$

$$\boxed{v\frac{dv}{dx} = g/3 = a}$$



Loss in P.E

$$= (\rho L)\, g\, \frac{L}{2}$$

$$= \frac{\rho g L^2}{2}$$

gain in k.E $= \frac{1}{\cancel{2}}(\rho L)\, \frac{\cancel{2} g L}{3}$

$$= \frac{\rho g L^2}{3}$$

L

$$\frac{\rho g L^2}{2} = \frac{\rho g L^2}{3} + Q$$

$$Q = \rho g L^2 \left\{ \frac{1}{2} - \frac{1}{3} \right\} = \frac{\rho g L^2}{6}$$

IMT

$$\cancel{f}xg\,dt = \left\{\cancel{f}(x+dx)(v+dv)\right\} - \left\{\cancel{f}xv\right\}$$

$$xg\,dt = \cancel{xv} + \underline{x\,dv + v\,dx} + dx\,dv - \cancel{xv}$$

$$dx\,dv \to 0$$

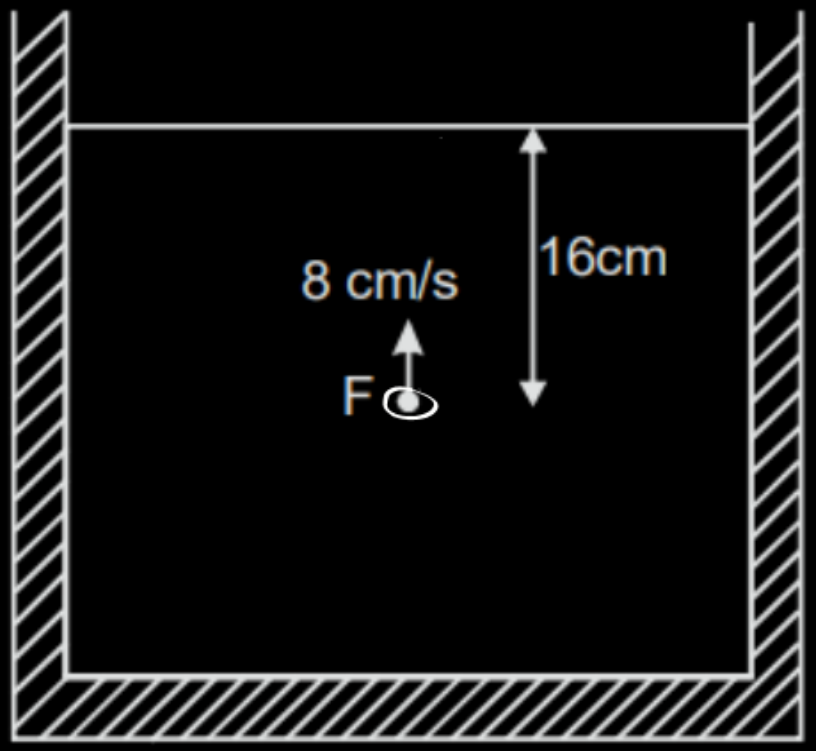
$$(vx)\,xg\,dt = (vx)\,d(vx)$$

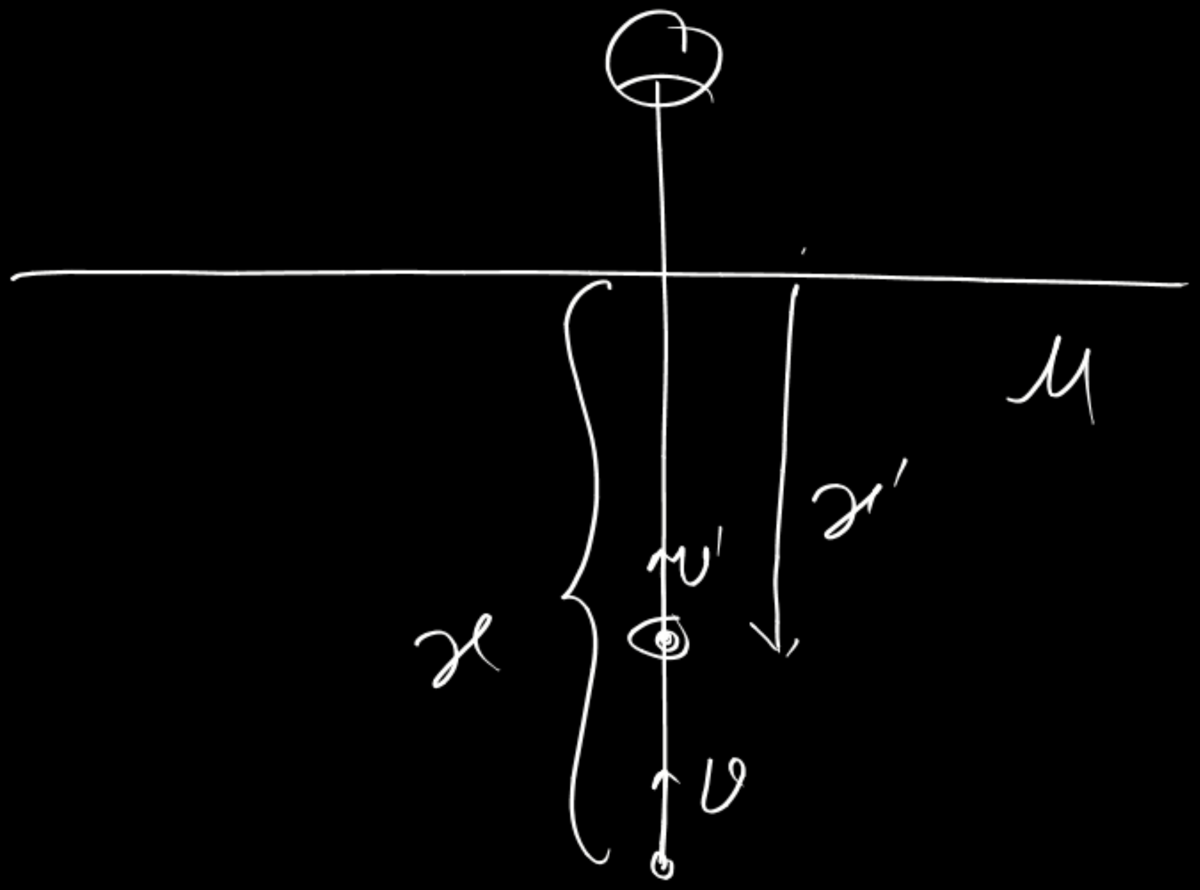
$$\int_0^x g x^2\,dx = \int_0^{vx} vx\,d(vx) \qquad \int y\,dy \quad \frac{y^2}{2}$$

$$\boxed{v = \sqrt{\frac{2gx}{3}}}$$

$$g\left(\frac{x^3}{3}\right)_0^x = \left(\frac{(vx)^2}{2}\right)_0^{vx} \Rightarrow g\frac{x^3}{3} = \frac{v^2x^2}{2}$$

Consider the situation in figure. A small fish F is moving in vertically upward diection with velocity +8 cm/s. If the time dependent refractive index of water is $\mu(t) = \frac{4}{3} + \frac{t}{3}$; (t is in sec.) then velocity (in cm/sec) of fish as seen from above at t = 0 sec is. (Consider upward direction as positive)





$$x' = \frac{x}{\mu}$$

$$= \frac{3x}{(4+t)}$$

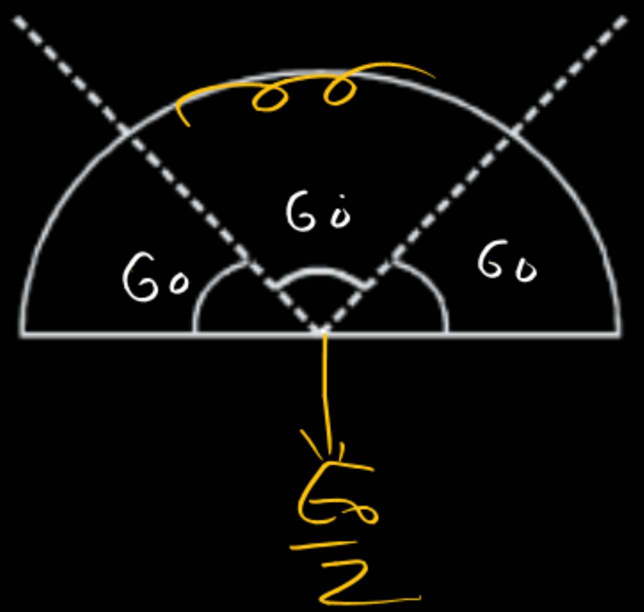
$$v' = \frac{dx'}{dt} = 3\,\frac{(4+t)\left(\frac{dx}{dt}\right) - x \cdot 1}{(4+t)^2}$$

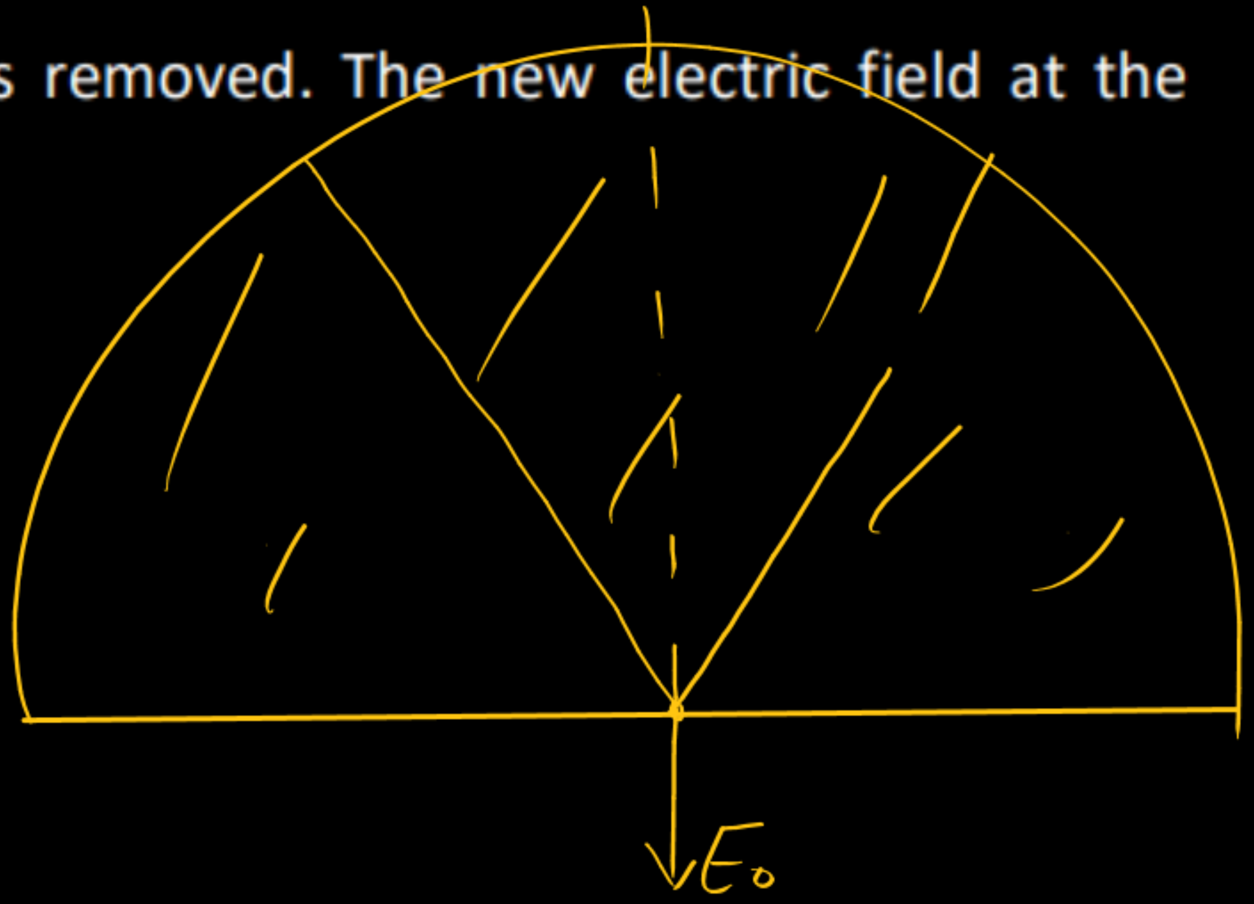
$$= \frac{3(4+t)(-8) - 3x}{(4+t)^2} = \frac{12(-8) - 3\times 16}{16} = -6 - 3$$

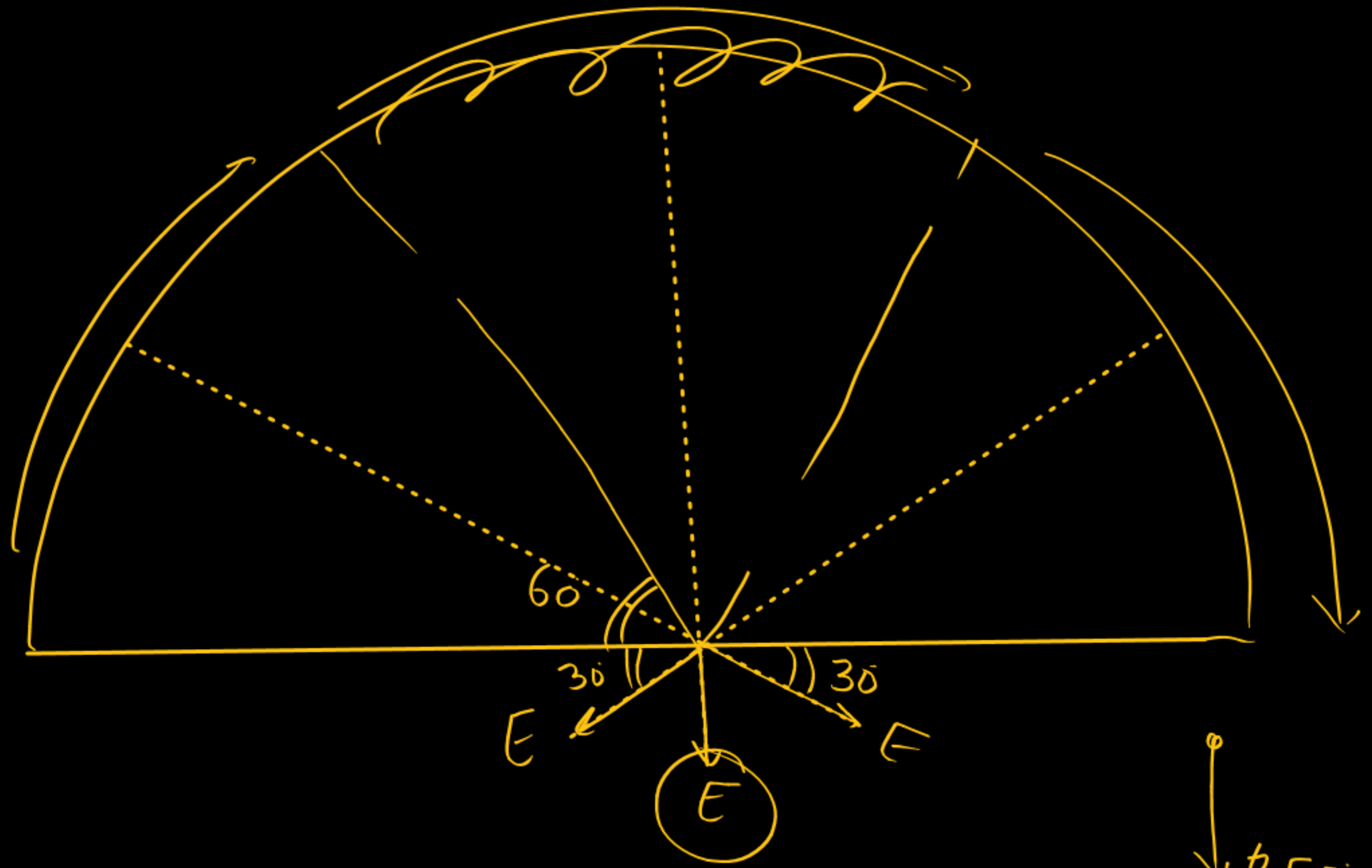
$$= \cancel{-7}\ -9 \text{ cm/sec}$$

A solid hemisphere is having a uniform volume charge density. Electric field at its centre is E_0. It is cut into three identical parts as shown and the central part is removed. The new electric field at the centre $\frac{E_0}{x}$. Then x is :

2







$$E_{net} = \cancel{2}E\sin\cancel{30}$$

$$= E = \frac{E_0}{2}$$

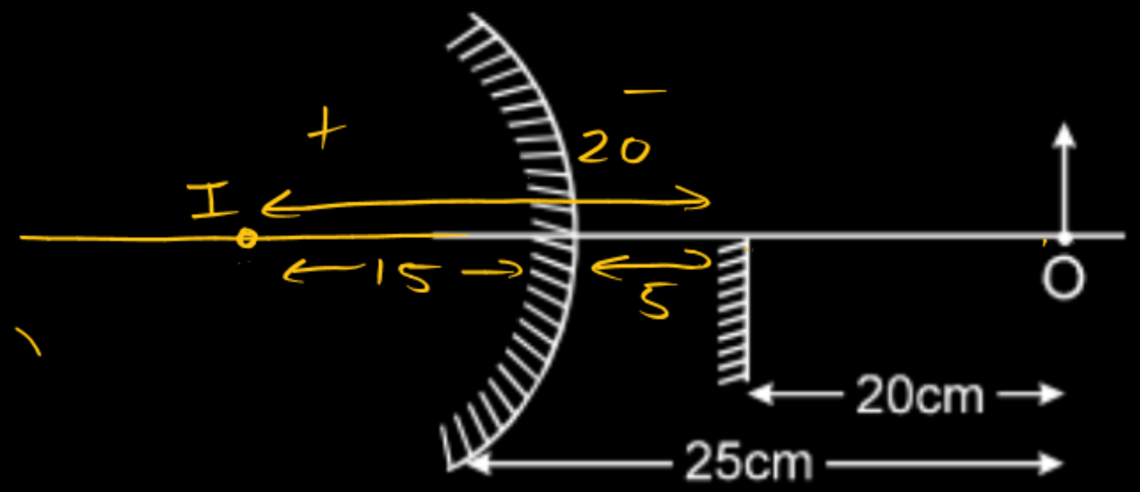
$$\cancel{2}E\sin\cancel{30} +$$

$$+ E$$

$$\boxed{2E = E_0}$$

$$E = \frac{E_0}{2}$$

In the figure, an object is placed at distance 25 cm from the surface of a convex mirror, and a plane mirror is set so that the image formed by the two mirrors lie adjacent to each other in the same plane. The plane mirror is placed at 20 cm from the object. The radius of curvature of the convex mirror is R (in cm). Then value of $\frac{R}{15}$



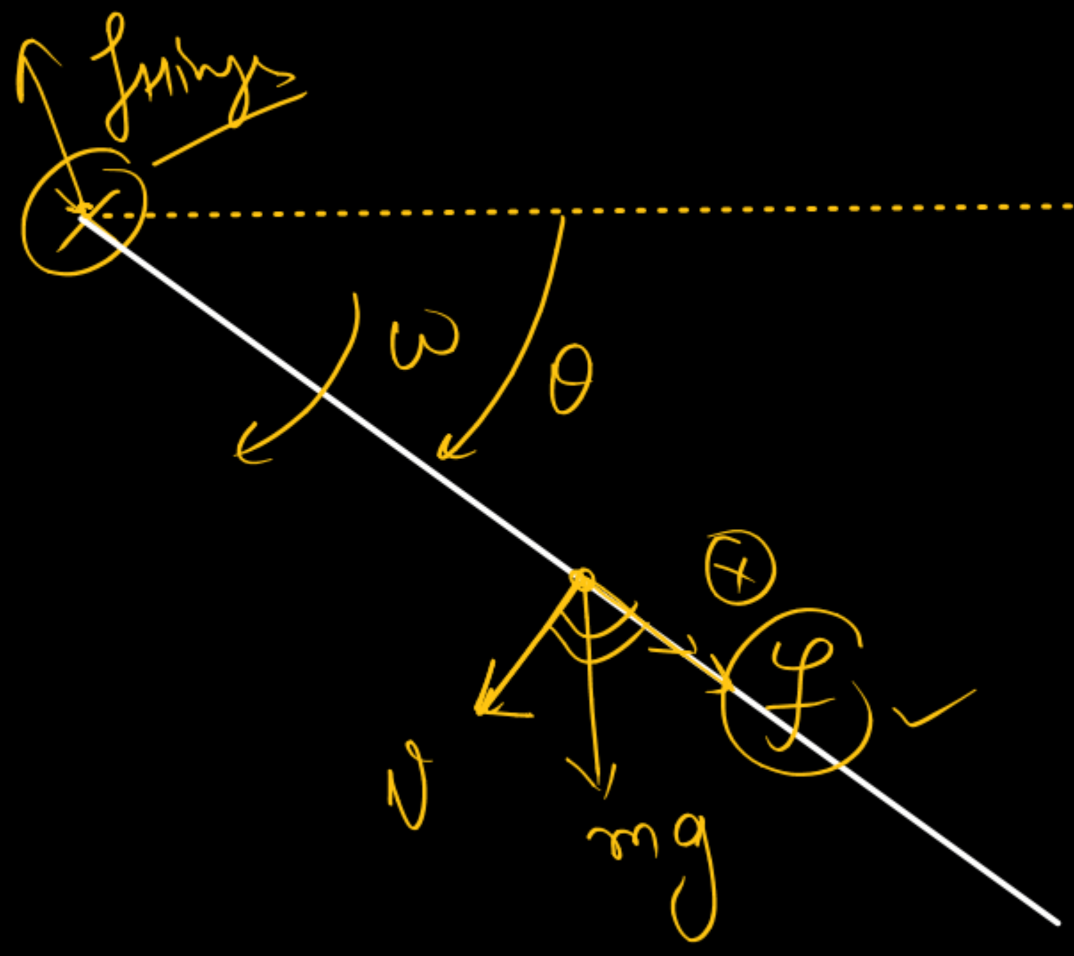
$$\frac{R}{15} = \frac{10}{2} = 5$$

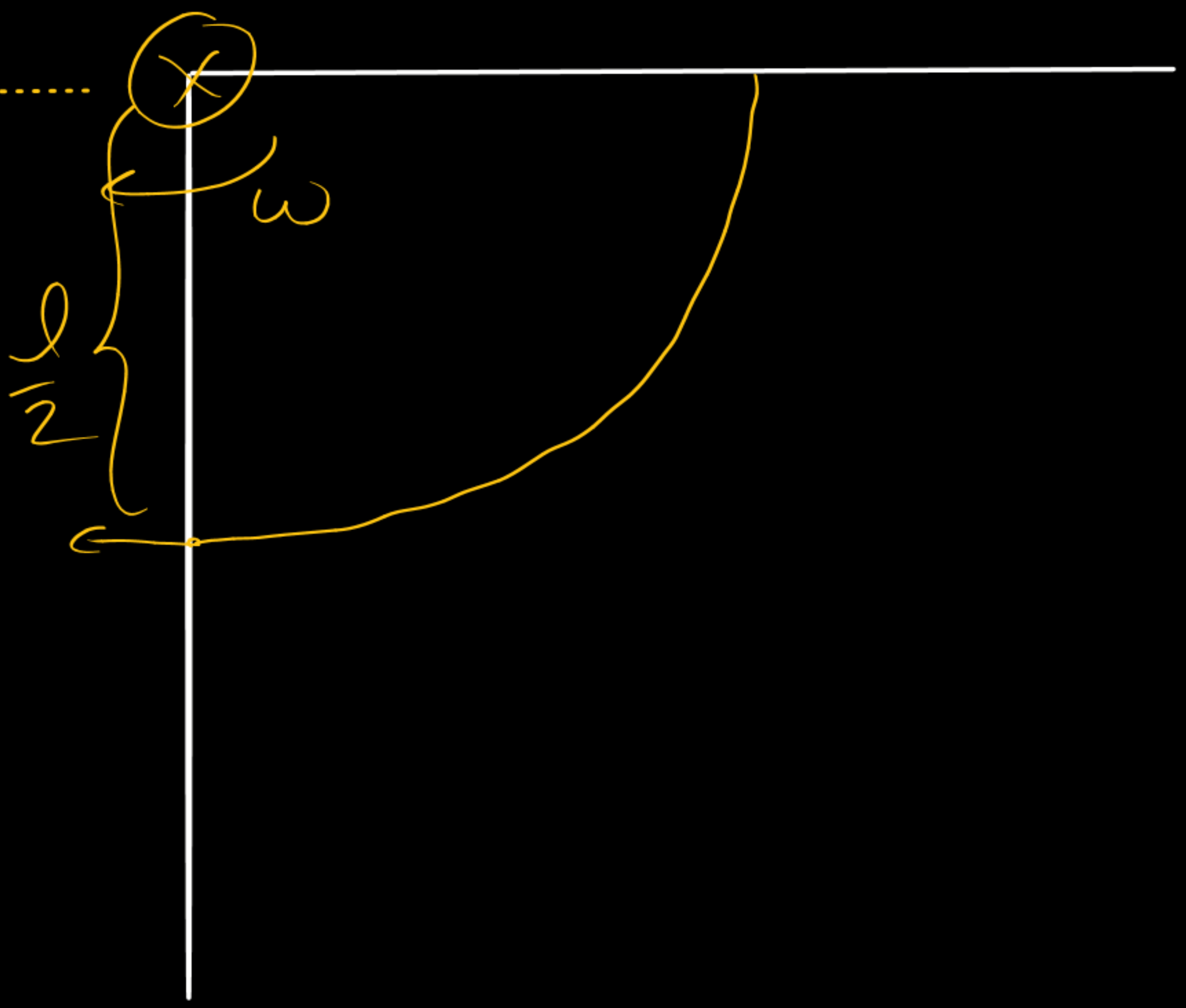
$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{+15} + \frac{1}{-25} = \frac{1}{f} = \frac{10-6}{150} \Rightarrow f = \frac{150}{4} \qquad R = \frac{150}{2}\text{ cm}$$

A non conducting rod of length 2m (ℓ) is hinged at one end and a charge of $\frac{1}{10}$ C (q) is distributed uniformly over it. At t = 0, it is released from the position shown. There exist a uniform magnetic field of $\sqrt{15}$ T (B) inside the plane of motion of rod. If mass of rod is 100 g (m), then find the value of hinge force (in N) when rod is rotated by $\frac{\pi}{2}$ due to gravity.

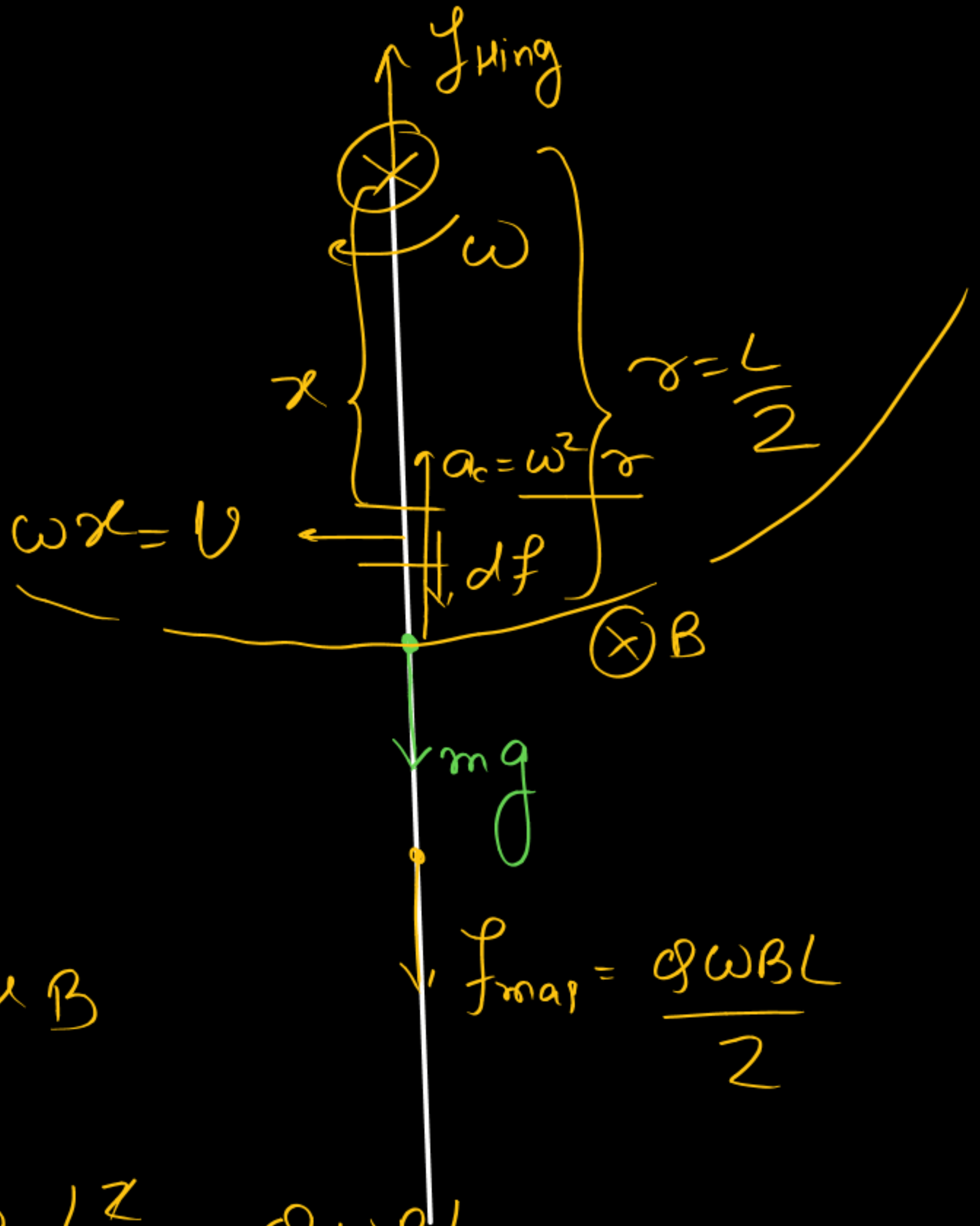
4





$$mg\frac{l}{\cancel{2}} = \frac{1}{\cancel{2}}\left(\frac{\cancel{m}l^{\cancel{2}}}{3}\right)\omega^2 - 0$$

$$\boxed{\omega = \sqrt{\frac{3g}{l}}}$$



$q\vec{v} \times \vec{B}$

$$df = \left(\frac{q}{L}dx\right)\omega x B$$

$$f = \int df = \frac{q\omega B}{\cancel{L}}\frac{L^{\cancel{2}}}{2} = \frac{q\omega BL}{2}$$

$$f_H - mg - \frac{q\omega BL}{2} = m\omega^2 \frac{L}{2}$$

$$f_H = mg + \frac{q\omega BL}{2} + m\frac{3g}{\cancel{L}} \cdot \frac{\cancel{L}}{2}$$

$$= \frac{5mg}{2} + \frac{q\omega BL}{2}$$

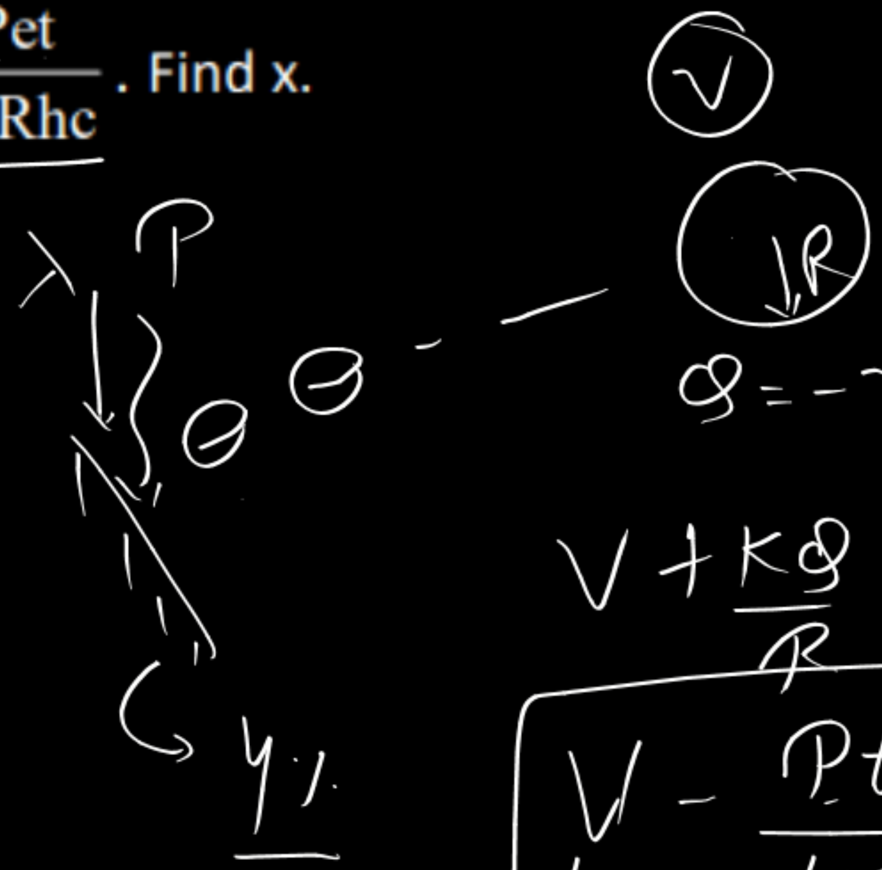
$$= \frac{5}{2} \times 0.1 \times 10 + \frac{\frac{1}{10} \times \sqrt{15}\sqrt{\frac{30}{2}} \times 2}{2}$$

15

$$= 2.5 + 1.5$$

$$= 4 \text{ Newton}$$

The work function of a certain metal is $\frac{hc}{\lambda_0}$. When a monochromatic light of wavelength $\lambda < \lambda_0$ is incident suchthat the plate gains a total power P. If the efficiency of photoelectric emission is η% and all the emitted photoelectrons are captured by a hollow conducting sphere of radius R already charged to potential V, then neglecting anyinteraction between plate and the sphere, expression of potential of the sphere at time t is $V + \frac{\eta \lambda P e t}{x\, \pi \varepsilon_0 R h c}$. Find x.



$$q = -n_e e$$

$$V + \frac{kq}{R}$$

$$\boxed{V - \frac{Pt\lambda y e}{4\pi\varepsilon_0 100 hc R}}$$

$$Pt = n\frac{hc}{\lambda}$$

$$n = \frac{P\lambda t}{hc}$$

$$\boxed{x = -400}$$

$$\frac{n_e}{n} \times 100 = y$$

$$n_e = \frac{ny}{100}$$

$$= \frac{Pt\lambda y}{hc\, 100}$$